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# The Riemann-Hurwitz formula and an application to the polynomial Catalan conjecture

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Bachelor's Thesis

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September, 2021

# Contents

# 1 Introduction

Eugène Charles Catalan conjectured in 1844 that $2^3$ and $3^2$ are the only consecutive powers of natural numbers. The conjecture could be proven only in 2002 by Preda Mihăilescu. Further information about Catalan's conjecture and Mihăilescu's proof can be found in [M04]. Instead of looking at powers of natural numbers one can also study powers of coprime polynomials, where the goal is to find the smallest possible degree of the difference. In 1965 the mathematicians Birch, Chowla, Hall and Schinzel made the following conjecture in [BCHS65].

**Conjecture 1.1** (Polynomial Catalan conjecture)**.** *For coprime polynomials $A, B \in \mathbb{C}[X]$ with degrees $\deg(A) = 2k$ and $\deg(B) = 3k$ the difference $R = A^3 - B^2$ has degree $\deg(R) \geq k + 1$. Furthermore, this bound is attained for infinitely many values of $k$.*

The lower bound was proven by Davenport in the same year. The attainability statement took longer to prove, in 1981 W. Stothers could show in [S81] that the lower bound is attained for all values of $k$. We will give a proof of the polynomial Catalan Conjecture following the idea of N. Adrianov, F. Pakovich and A. Zvonkin from the book [APZ20]. The approach of their proof is to relate the problem of the polynomial Catalan conjecture to the theory of holomorphic maps between Riemann surfaces. Riemann studied these surfaces in the 19th century, the most concise definition of a Riemann surface is that it is a connected, 1-dimensional complex manifold. One can define holomorphic maps between these surfaces and an important result called the Riemann-Hurwitz formula relates properties of a holomorphic map between Riemann surfaces with their genera. Riemann already used a special case of the formula, although a proof was given only later for example by Hurwitz. An exposition of the Riemann-Hurwitz formula can be found in [O16]. This formula will also be the central tool in our proof of the lower bound of the polynomial Catalan conjecture. To show the attainability we will work with dessins d'enfants. These are plane graphs arising from certain holomorphic functions between the Riemann sphere and itself. Since they encode important information about their corresponding holomorphic function, we will be able to shift the study of holomorphic functions to the study of their dessins. Dessins d'enfants were first introduced in 1984 by A. Grothendieck in the *Esquisse d'un programme* and are currently an active area of research.

In Section 2 we will introduce Riemann surfaces and holomorphic maps between them. Furthermore we define the needed concepts for the subsequent sections, such as ramification, degree, and genus. To define the genus, we will also briefly touch upon cohomology and homology. A part will also be

dedicated to triangulations of Riemann surfaces, since they are needed in the proof of the Riemann-Hurwitz formula.
Section 3 states four fundamental theorems concerning Riemann surfaces, including the Riemann-Hurwitz formula. Since only the Riemann-Hurwitz formula will be directly applied in the last chapter, it will be the only theorem in that section for which we give a proof.
Finally, in Section 4 we will work towards proving the polynomial Catalan conjecture.

I want to thank my advisor Prof. Dr. Peter Jossen who was very helpful in answering all of my questions and providing me with an interesting topic. Furthermore, I want to thank Vanessa Häusermann and Lukas Oestmann for carefully proofreading the thesis, and Ruilong Wäckerlin for giving useful feedback on parts of the thesis.

# 2 Riemann surfaces

## 2.1 Riemann surfaces and maps

We start by introducing the concept of a Riemann surface, and giving some examples. We will define holomorphic maps between Riemann surfaces and show some properties about them that will be needed later. As a general reference for this section several textbooks about Riemann Surfaces were used, namely [D11, Chapters 3,4, 7], [F81, Chapters 1,2], [FK92, Chapter I], [M95, Chapters I-III], [S14, Chapter 4]. If nothing specifically stated, the proofs we present here follow the proofs that can be found in these textbooks.

**Definition 2.1.** A *Riemann surface* is a connected, 1-dimensional complex manifold.

To be more explicit, a Riemann surface $M$ is a Hausdorff topological space with a family of open subsets $\{U_\alpha\}_{\alpha\in I}$ for some index set $I$, such that this family covers the space $M$, that is,

$$\bigcup_{\alpha\in I} U_\alpha = M.$$

Additionally, for each $\alpha \in I$ there is a homeomorphism $z_\alpha : U_\alpha \to V_\alpha$, where $V_\alpha$ is an open subset of the complex plane $\mathbb{C}$. These homeomorphisms have to satisfy that $\forall \alpha, \beta \in I$ the transition map

$$z_\alpha \circ z_\beta^{-1} : z_\beta(U_\alpha \cap U_\beta) \to z_\alpha(U_\alpha \cap U_\beta)$$

is holomorphic.

**Definition 2.2.** For a Riemann surface $M$, the pair $(U_\alpha, z_\alpha)$ as described above is called a *chart* or *local coordinate* of $M$ and the collection of all charts $\{(U_\alpha, z_\alpha)\}_{\alpha\in I}$ is called an *atlas* of $M$.

From the definition it is clear that an open connected subset $U \subset \mathbb{C}$ is a Riemann surface. Indeed, $U$ is a Hausdorff space as a subset of $\mathbb{C}$ and an atlas is given by the chart $(U, \mathrm{id}|_U)$. Furthermore, every open and connected subset of a Riemann surface is again a Riemann surface by restricting the charts.

**Example 2.3.** An important example of a Riemann surface is the *Riemann sphere* $\mathbb{P}^1\mathbb{C}$. As a topological space, it is the one-point compactification of $\mathbb{C}$. The one-point compactification is defined to be the topological space $\mathbb{P}^1\mathbb{C} = \mathbb{C} \cup \{\infty\}$, where $\infty$ denotes an element not already contained in $\mathbb{C}$. The open sets of the compactification are given by the open subsets $U \subset \mathbb{C}$ together with the sets $U \subset \mathbb{P}^1\mathbb{C}$ containing the point $\infty$ and such that $\mathbb{P}^1\mathbb{C} \setminus U \subset \mathbb{C}$ is compact in $\mathbb{C}$. This topological space is Hausdorff and we will give an explicit atlas to show that $\mathbb{P}^1\mathbb{C}$ is a Riemann surface. The pair $(U_1, z_1) = (\mathbb{C}, id)$ is a chart with the identity map $id : \mathbb{C} \to \mathbb{C}$. The second chart in the atlas is $(U_2, z_2) = (\mathbb{P}^1\mathbb{C} \setminus \{0\}, z_2)$, where the map $z_2$ is given by

$$z_2 : \mathbb{P}^1\mathbb{C} \setminus \{0\} \to \mathbb{C},\ x \mapsto \frac{1}{x}$$

with the convention $\frac{1}{\infty} = 0$. We claim that the set $\{(U_1, z_1), (U_2, z_2)\}$ is an atlas. The two transition maps coincide

$$z_1 \circ z_2^{-1} = z_2 \circ z_1^{-1} : \mathbb{C} \setminus \{0\} \to \mathbb{C} \setminus \{0\},\ x \mapsto \frac{1}{x}.$$

We see that the transition maps are holomorphic functions because 0 is not in their domain, which shows that we have given the Riemann sphere the structure of a Riemann surface.
The name Riemann sphere can be explained because the Riemann sphere is homeomorphic to the 2-dimensional sphere $S^2 \in \mathbb{R}^3$ as topological spaces.
The notation $\mathbb{P}^1\mathbb{C}$ denotes the complex projective line, which is also homeomorphic to the one-point compactification of the complex plane.

Before we introduce holomorphic maps between Riemann surfaces, we will first view them as real 2-manifolds with smooth transition maps. This allows us to study whether a Riemann surface is orientable or not, recall the following definition.

**Definition 2.4.** A real $n$-dimensional manifold is called *orientable* if it has an atlas where the Jacobian of every transition map has positive determinant at every point in its domain. The $n$-manifold together with such an atlas is called *oriented*.

Because Riemann surfaces satisfy the additional property that their transition maps are holomorphic, we can determine the question about their orientability in general.

**Proposition 2.5.** *Riemann surfaces are oriented.*

*Proof.* We follow the proof given in Chapter 1.2 of [G89]. Let $M$ be a Riemann surface with atlas $\{(U_\alpha, z_\alpha)\}_{\alpha \in I}$. For any $\alpha, \beta \in I$ the transition map

$$z_{\alpha\beta} = z_\alpha \circ z_\beta^{-1} : z_\beta(U_\alpha \cap U_\beta) \to z_\alpha(U_\alpha \cap U_\beta)$$

can be written as $z_{\alpha\beta}(x + iy) = u(x, y) + iv(x, y)$ where $x + iy \in z_\beta(U_\alpha \cap U_\beta)$ is any point, and $u$ and $v$ are smooth real-valued functions. This allows us to view the Riemann surface as a 2-manifold, so we can check the definition of orientability. The Jacobian $J_{z_{\alpha\beta}}(x_0, y_0)$ at the point $(x_0, y_0) \in z_\beta(U_\alpha \cap U_\beta)$ of the function $z_{\alpha\beta}$ viewed as a smooth function between open subset in $\mathbb{R}^2$ as described above is

$$J_{z_{\alpha\beta}}(x_0, y_0) = \begin{pmatrix} \frac{\partial u}{\partial x}(x_0, y_0) & \frac{\partial u}{\partial y}(x_0, y_0) \\ \frac{\partial v}{\partial x}(x_0, y_0) & \frac{\partial v}{\partial y}(x_0, y_0) \end{pmatrix}.$$

The determinant is

$$\det(J_{z_{\alpha\beta}}(x_0, y_0)) = \frac{\partial u}{\partial x}(x_0, y_0) \cdot \frac{\partial v}{\partial y}(x_0, y_0) - \frac{\partial u}{\partial y}(x_0, y_0) \cdot \frac{\partial v}{\partial x}(x_0, y_0).$$

Since $z_{\alpha\beta}$ is holomorphic, the functions $u$ and $v$ satisfy the Cauchy-Riemann equations

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x}. \end{aligned}$$

Substituting the Cauchy-Riemann equations in the expression for the determinant of the Jacobian gives

$$\det(J_{z_{\alpha\beta}}(x_0, y_0)) = \left(\frac{\partial u}{\partial x}(x_0, y_0)\right)^2 + \left(\frac{\partial u}{\partial y}(x_0, y_0)\right)^2.$$

Thus $\det(J_{z_{\alpha\beta}}(x_0, y_0)) > 0$ at every point $(x_0, y_0)$ in the domain, which shows that $M$ is oriented. $\square$

**Definition 2.6.** Let $f : M \to N$ be a continuous map between Riemann surfaces $M$ and $N$. The map $f$ is called *holomorphic at a point* $p \in M$ if there exists charts $(U, z)$ of $M$ with $p \in U$, and $(V, w)$ of $N$ with $f(p) \in V$ such that the composition $w \circ f \circ z^{-1}$ is holomorphic at $z(p)$.

**Definition 2.7.** A continuous map $f : M \to N$ between Riemann surfaces is called *holomorphic* if for every local coordinate $(U, z)$ of $M$ and every local coordinate $(V, w)$ on $N$ the composition

$$w \circ f \circ z^{-1} : z(U \cap f^{-1}(V)) \to w(V)$$

is holomorphic.

After having introduced the above definitions for maps between Riemann surfaces, we want to see how they relate to each other. Thus consider the following proposition.

**Proposition 2.8.** *Let $f : M \to N$ be a continuous map between Riemann surfaces. The map $f$ is holomorphic if and only if it is holomorphic at every point $p \in M$.*

*Proof.* We use the idea from the proof of [M95, Lemma 1.2.]. First assume that $f$ is a holomorphic map, then it is clearly holomorphic at every point $p \in M$.
Conversely, assume that $f$ is holomorphic at every point $p \in M$. Let $(U, z)$ and $(V, w)$ be arbitrary charts of $M$ and $N$ respectively and fix any point $p_0 \in U \cap f^{-1}(V)$. By assumption the map $f$ is holomorphic at $p_0$, thus there exists a chart $(U_0, z_0)$ of $M$ with $p_0 \in U_0$, and a chart $(V_0, w_0)$ of $N$ with $f(p_0) \in V_0$ such that the composition $w_0 \circ f \circ z_0^{-1}$ is holomorphic at $z_0(p_0)$. The charts $(U, z), (V, w)$ and the point $p_0$ were arbitrary, so to finish the prove of the proposition it is enough to show that $w \circ f \circ z^{-1}$ is holomorphic at $z(p_0)$. We can rewrite the composition

$$\begin{aligned} w \circ f \circ z^{-1} &= w \circ (w_0^{-1} \circ w_0) \circ f \circ (z_0^{-1} \circ z_0) \circ z^{-1} \\ &= (w \circ w_0^{-1}) \circ (w_0 \circ f \circ z_0^{-1}) \circ (z_0 \circ z^{-1}) \end{aligned}$$

with $z(p_0)$ in its domain. Note that $w \circ w_0^{-1}$ and $z_0 \circ z^{-1}$ are transition maps of the Riemann surfaces $M$, respectively $N$ and thus they are holomorphic. The term $w_0 \circ f \circ z_0^{-1}$ in the middle is holomorphic at $z_0(p_0)$ by assumption. Hence the composition is also holomorphic at $z(p_0)$. $\square$

**Example 2.9.** We will give an example of a holomorphic map between Riemann surfaces. Let $M = N = \mathbb{P}^1\mathbb{C}$ be the Riemann sphere from Example 2.3, using the same atlas. Consider an arbitrary, non-constant polynomial $P = \sum_{i=0}^{n} a_i X^i \in \mathbb{C}[X]$ with $a_n \neq 0$. This polynomial can be viewed as a continuous map $f : M \to N$ where we set $f(\infty) = \infty$ and $\forall x \in \mathbb{C}\ f(x) = \sum_{i=0}^{n} a_i x^i$. To verify that $f$ is a holomorphic map we will use the equivalence from Proposition 2.8 above. First, we check that $f$ is holomorphic at points $p \in \mathbb{C}$. Since the image $f(\mathbb{C}) = \mathbb{C}$, we can use the chart $(U_1, z_1)$. The composition $z_1 \circ f \circ z_1^{-1}$ becomes

$$f : \mathbb{C} \to \mathbb{C},\ x \mapsto f(x) = a_0 + a_1 x + ... + a_n x^n$$

as both charts are the identity function. Clearly, this is a holomorphic function on $\mathbb{C}$, so we have shown that $f$ is holomorphic at every point $p \in \mathbb{C} \subset \mathbb{P}^1\mathbb{C}$. It is only left to show that $f$ is holomorphic at $\infty \in \mathbb{P}^1\mathbb{C}$. Since $f(\infty) = \infty$, we need to use the chart $(U_2, z_2)$. The composition is

$$z_2 \circ f \circ z_2^{-1} : z_2(\mathbb{P}^1\mathbb{C} \setminus (\{0\} \cup f^{-1}(0))) \to \mathbb{C},$$
$$x \mapsto \frac{1}{f(1/x)} = \frac{x^n}{a_0 x^n + a_1 x^{n-1} + ... + a_n}.$$

At the point $z_2(\infty) = 0$, the above composition is the quotient of two holomorphic maps where the denominator does not vanish because $a_n \neq 0$ by assumption. Hence the composition is holomorphic at 0, and $f$ is holomorphic at $\infty$. Altogether, this shows that $f : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C}$ is a holomorphic map.

Note that because the definition of holomorphic maps between Riemann surfaces uses holomorphicity only locally, all of the local properties of holomorphic functions on open sets in the complex plane also hold for holomorphic maps between Riemann surfaces. We will see some examples of this below with the Identity Theorem and the Open Mapping Theorem.

**Proposition 2.10** (Identity Theorem). *Let $f, g : M \to N$ be holomorphic maps between Riemann surfaces. If there exists a set $A \subset M$ with a limit point $a \in M$ such that the restrictions satisfy $f|_A \equiv g|_A$, then $f \equiv g$ identically on $M$.*

*Proof.* We follow the proof of [F81, Theorem 1.11.]. Consider the set

$$D = \{p \in M \mid \exists W \subset M \text{ an open neighbourhood of } p \text{ such that } f|_W \equiv g|_W\}.$$

From the definition, it is clear that $D \subset M$ is an open subset, and we will show that it is also closed. Let $p \in \partial D$ be any point in the topological boundary of $D$. From the continuity of $f$ and $g$ it follows that $f(p) = g(p)$. By restricting charts if necessary, we can choose a chart $(U, z)$ of $M$ with $p \in U$ and a chart $(V, w)$ of $N$ such that $f(U) \subset V$ and $g(U) \subset V$. Consider the compositions

$$\tilde{f} = w \circ f \circ z^{-1} : z(U) \to w(V)$$
$$\tilde{g} = w \circ g \circ z^{-1} : z(U) \to w(V).$$

Since $\tilde{f} \equiv \tilde{g}$ on $z(U \cap D) \neq \varnothing$ and the set $z(U \cap D)$ has a limit point $z(p) \in z(U)$, we can use the Identity Theorem on the complex plane to derive that $\tilde{f} \equiv \tilde{g}$ identically on $z(U)$. This implies $f \equiv g$ on $U$ as both $z$ and $w$ are homeomorphisms. We have shown that on the neighbourhood $U$ of $p$, the maps $f$ and $g$ are identical, hence $p \in D$. Because $p$ was an arbitrary point of the boundary of $D$, it follows that $D$ equals its closure and hence is closed

itself.
We claim that $D \neq \varnothing$, in particular that the point $a \in D$. This can be shown by choosing charts $(U', z')$ of $M$ and $(V', w')$ of $N$ such that $a \in U'$, and $f(U') \subset V'$, $g(U') \subset V'$. Applying the Identity theorem for holomorphic functions in $\mathbb{C}$ for the compositions $w' \circ f \circ (z')^{-1}$ and $w' \circ g \circ (z')^{-1}$ as above shows that $a \in D$. To conclude, we have shown that $D \subset M$ is a non-empty set that is open and closed. Hence by the connectedness of $M$ it follows $M = D$. □

**Remark 2.11.** For a non-constant holomorphic map $f : M \to N$ between Riemann surfaces, the preimage $f^{-1}(q)$ for any point $q \in N$ is discrete. Indeed, if the preimage $f^{-1}(q)$ were not discrete, by the Identity Theorem it would follow that $f$ is identical to the constant map $c_q : M \to N, p \mapsto q$. This is a contradiction to the assumption that $f$ is not constant. In particular, if $M$ is compact the preimage $f^{-1}(q)$ is finite, as a closed discrete subset of a compact space is finite.

**Proposition 2.12** (Open Mapping Theorem)**.** *Any non-constant holomorphic map $f : M \to N$ between Riemann surfaces is an open map, that is, for any open subset $U \subset M$, the image $f(U) \subset N$ is open.*

*Proof.* Let $U \subset M$ be an open set, let $\{(U_\alpha, z_\alpha)\}_{\alpha \in I}$ denote the atlas of $M$, and $\{V_\beta, w_\beta)\}_{\beta \in J}$ the atlas of $N$. The image $f(U)$ can be written as

$$f(U) = f\left(\bigcup_{\alpha \in I} (U \cap U_\alpha)\right) = \bigcup_{\alpha \in I} f(U \cap U_\alpha).$$

We show that for any $\alpha \in I$, the image $f(U \cap U_\alpha)$ is open in $N$. This finishes the proof as it implies $f(U) \subset N$ is open as a union of open sets. Fix any $\alpha \in I$, since the $\{V_\beta\}_{\beta \in J}$ cover $N$, we have

$$U \cap U_\alpha = \bigcup_{\beta \in J} (U \cap U_\alpha \cap f^{-1}(V_\beta)).$$

For any $\beta \in J$ we can consider the composition

$$\varphi = w_\beta \circ f \circ z_\alpha^{-1} : z_\alpha(U_\alpha \cap f^{-1}(V_\beta)) \to w_\beta(V_\beta).$$

This is a non-constant holomorphic map on $\mathbb{C}$, because if $f$ were locally constant on $U_\alpha \cap f^{-1}(V_\beta)$, then $f$ would be globally constant by the Identity Theorem. We can apply the Open Mapping Theorem in $\mathbb{C}$ for the map $\varphi$ and it follows that $\varphi(z_\alpha(U \cap U_\alpha \cap f^{-1}(V_\beta)))$ is open in $w_\beta(V_\beta)$. Since both $z_\alpha$ and $w_\beta$ are homeomorphisms, this implies $f(U \cap U_\alpha \cap f^{-1}(V_\beta)) \subset V_\beta$ is open. As a chart, the set $V_\beta \subset N$ is open, thus $f(U \cap U_\alpha \cap f^{-1}(V_\beta)) \subset N$ is an open subset of $N$. This holds for any $\beta \in J$, so each

$$f(U \cap U_\alpha) = \bigcup_{\beta \in J} f(U \cap U_\alpha \cap f^{-1}(V_\beta))$$

is open as a union of open subsets. $\square$

**Proposition 2.13.** *Let $f : M \to N$ be a non-constant holomorphic map between Riemann surfaces, where $M$ is compact and non-empty. Then the map $f$ is surjective.*

*Proof.* By the Open Mapping Theorem, the image $f(M) \subset N$ is open. Because $f(M)$ is the image of a compact set under a continuous map, it is compact. As any compact subset of a Hausdorff space is closed, it follows $f(M) \subset N$ is closed. We conclude that $f(M) = N$, since $N$ is connected and $f(M)$ is a non-empty subset that is both open and closed. $\square$

**Remark 2.14.** By Proposition 2.13 above, a holomorphic, non-constant map $f : M \to N$ between Riemann surfaces where $M$ is compact is surjective. In particular, $N$ is the image of a compact space under a continuous function, hence $N$ is also compact. This further implies that every holomorphic map from a compact Riemann surface to a non-compact Riemann surface must be constant.

**Definition 2.15.** The Riemann surfaces $M$ and $N$ are said to be *isomorphic as Riemann Surfaces* or *holomorphically isomorphic* if there exists a map $f : M \to N$ that is a holomorphic bijection with holomorphic inverse, that is, $f$ is biholomorphic.

**Example 2.16.** Another important example of Riemann surfaces are *elliptic curves*. To construct an elliptic curve, consider two complex numbers $\omega_1, \omega_2 \in \mathbb{C}$ that are linearly independent over $\mathbb{R}$. They define a lattice

$$\Gamma = \omega_1\mathbb{Z} + \omega_2\mathbb{Z} = \{\omega_1 n + \omega_2 m \mid n, m \in \mathbb{Z}\} \subset \mathbb{C}.$$

We consider an equivalence relation on the complex numbers $\mathbb{C}$, where $z, x \in \mathbb{C}$ are equivalent if and only if their difference satisfies $z - x \in \Gamma$. The set of equivalence classes is denoted $\mathbb{C}/\Gamma$ and we equip it with the quotient topology. Recall that in the quotient topology a set $U \subset \mathbb{C}/\Gamma$ is open if and only if $\pi^{-1}(U) \subset \mathbb{C}$ is open, where $\pi : \mathbb{C} \to \mathbb{C}/\Gamma$ denotes the projection. Since the projection $\pi$ is a continuous and surjective map, and $\mathbb{C}$ is connected, it follows that the image $\mathbb{C}/\Gamma$ of the projection map is also connected. One can verify that the topological space $\mathbb{C}/\Gamma$ is Hausdorff. Note that $\pi$ is an open map, for if $V \subset \mathbb{C}$ is an open subset we have that

$$\pi^{-1}(\pi(V)) = \bigcup_{\omega \in \Gamma} \{v + \omega \mid v \in V\}$$

which is open as it is a union of open subsets in $\mathbb{C}$. By the definition of the quotient topology this shows that $\pi(V) \subset \mathbb{C}/\Gamma$ is open.
To show that $\mathbb{C}/\Gamma$ is a Riemann surface, the only thing left is to give an atlas. Consider an open subset $V \subset \mathbb{C}$ such that any two points $u, v \in V$ define different equivalence classes in the quotient $\mathbb{C}/\Gamma$. The image $\pi(V) \subset \mathbb{C}/\Gamma$ is open as we have shown above. We denote $U = \pi(V)$, the restriction map $\pi|_V : V \to U$ is bijective, continuous, and open because it is the restriction of an open map to an open subset $V \subset \mathbb{C}$. Hence it follows that $\pi|_V$ is a homeomorphism and thus $\varphi = (\pi|_V)^{-1} : U \to V$ is a chart of $\mathbb{C}/\Gamma$. Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in I}$ be all the charts we can construct in this way. Clearly the open sets $U_\alpha$ cover $\mathbb{C}/\Gamma$, we need to show that the transition maps are holomorphic. Let $\varphi_\alpha : U_\alpha \to V_\alpha$, $\varphi_\beta : U_\beta \to V_\beta$, $\alpha, \beta \in I$ be two charts and denote the composition

$$\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$$

by $\psi$. Note that for any $z \in \varphi_\beta(U_\alpha \cap U_\beta)$ it holds

$$\pi(\psi(z)) = \pi(\varphi_\alpha(\varphi_\beta^{-1}(z))) = \pi((\pi|_{V_\alpha})^{-1}((\pi|_{V_\beta})(z))) = \pi(z).$$

Hence $\psi(z)$ and $z$ are equivalent in the quotient $\mathbb{C}/\Gamma$, that is, $\psi(z) - z \in \Gamma$. This implies that the image of the map

$$\xi : \varphi_\beta(U_\alpha \cap U_\beta) \to \mathbb{C},\ z \mapsto \psi(z) - z$$

lies in the discrete subset $\Gamma \subset \mathbb{C}$. The map $\xi$ clearly is continuous and thus it must be constant on each connected component of $\varphi_\beta(U_\alpha \cap U_\beta)$. In particular $\xi$ is a holomorphic map, and thus the transition map $\psi$ is holomorphic as well.
One can further show all the elliptic curves that arise for some choice of $\mathbb{R}-$linearly independent $\omega_1, \omega_2 \in \mathbb{C}$ are homeomorphic as topological spaces to the standard torus $\mathbb{T} = S^1 \times S^1$. Thus we also call these Riemann surfaces complex tori. But it is important to note the difference that they are not all isomorphic as Riemann surfaces. In other words, there exists a homeomorphism between every two elliptic curves but this homeomorphism is not in general a biholomorphic map between them.

**Definition 2.17.** A holomorphic map from a Riemann surface $M$ into $\mathbb{C}$ is called a *holomorphic function* on $M$.

**Remark 2.18.** For any compact Riemann surface $M$, the holomorphic functions on $M$ are precisely the constant functions. This follows from Remark 2.14, as $\mathbb{C}$ is not compact.

We denote the set of all holomorphic functions on a Riemann surface $M$ by $\mathcal{O}(M)$. From the case of holomorphic functions on $\mathbb{C}$, one can see that for two holomorphic functions on $M$ their product and sum is again a holomorphic

function on $M$. With the constant functions with value 0 and 1, the set $\mathcal{O}(M)$ forms a ring.
As all the constant functions $M \to \mathbb{C}$ are clearly holomorphic on $M$, there exists a ring homomorphism $\mathbb{C} \to \mathcal{O}(M)$, that sends an element $z \in \mathbb{C}$ to the constant function with value $z$. Hence we can further view $\mathcal{O}(M)$ as a $\mathbb{C}$-algebra.
Instead of only considering the holomorphic functions on the whole Riemann surface $M$, we can assign to every open subset $U \subset M$ the ring $\mathcal{O}(U)$ of holomorphic functions on $U$. For any open sets $U \subset V \subset M$, the usual restriction of functions gives a ring homomorphism $\mathcal{O}(V) \to \mathcal{O}(U)$. After a verification one sees that $\mathcal{O}$ defines a sheaf of rings on the topological space $M$.

After holomorphic functions, we also want to introduce meromorphic functions on Riemann surfaces. In the following definition, a punctured neighbourhood of a point $p$ on a Riemann surface $M$ means a set $U \setminus \{p\}$, where $U \subset M$ is an open neighbourhood of $p$.

**Definition 2.19.** Consider a function $f : M \to \mathbb{C}$ on a Riemann surface $M$ that is holomorphic on a punctured neighbourhood of a point $p \in M$. The function $f$ has a *removable singularity* at $p \in M$ if there exists a chart $(U, z)$ of $M$ with $p \in U$, and such that the composition $f \circ z^{-1}$ has a removable singularity at $z(p)$ in the usual sense for complex valued functions on $\mathbb{C}$. Analogously, we define *poles* and *essential singularities* of $f$ at the point $p$.

**Definition 2.20.** A function $f : M \to \mathbb{C}$ on a Riemann surface $M$ is called *meromorphic* at the point $p \in M$ if it is holomorphic, has a pole, or has a removable singularity at $p$. Furthermore, the function $f$ is *meromorphic* on an open set $U \subset M$ if it is meromorphic at every point $p \in U$.

We denote the set of all meromorphic functions on a Riemann surface $M$ by $\mathcal{M}(M)$. After multiplying or adding two meromorphic functions we get another meromorphic function. The constant functions are holomorphic, hence in particular also meromorphic, and thus $\mathcal{M}(M)$ is also a $\mathbb{C}$-algebra. Analogous to meromorphic functions on $\mathbb{C}$, we can divide meromorphic functions in $\mathcal{M}(M)$ and get a meromorphic function on $M$ again when the denominator is not constantly 0 on $M$. This shows that $\mathcal{M}(M)$ is a field.

**2.21.** A meromorphic function $f : M \to \mathbb{C}$ on a Riemann surface $M$ can be viewed as a holomorphic map $\tilde{f} : M \to \mathbb{P}^1\mathbb{C}$. To realize this, we set $\tilde{f}(p) = \infty$ for any pole $p \in M$ of the function $f$, and remove all the removable singularities of $f$. It follows from the case of meromorphic functions on $\mathbb{C}$ that for any pole $p$ of $f$, the limit

$$\lim_{x \to p} |f(x)| = \infty.$$

This ensures that $\tilde{f}$ is continuous. We need to show that $\tilde{f}$ is also holomorphic on $M$. For any point $p \in M$ where $f$ is holomorphic, we can find a chart $(U, z)$ of $M$ with $p \in U$ and use the chart $(\mathbb{C}, id)$ of $\mathbb{P}^1\mathbb{C}$ as $\tilde{f}(p) \in \mathbb{C}$. The composition

$$id \circ \tilde{f} \circ z^{-1} = f \circ z^{-1} : z(U \cap \tilde{f}^{-1}(\mathbb{C})) \to \mathbb{C}$$

is holomorphic at $z(p)$ because $f$ is holomorphic at $p$.
This only leaves us to show that for a pole $p \in M$ of $f$, the map $\tilde{f}$ is holomorphic at $p$. Choose a chart $(U, z)$ of $M$ with $p \in U$. As a chart on $\mathbb{P}^1\mathbb{C}$, we need to choose $(V, w)$ where $V = \mathbb{P}^1\mathbb{C} \setminus \{0\}$ and $w : \mathbb{P}^1\mathbb{C} \setminus \{0\} \to \mathbb{C}, x \mapsto \frac{1}{x}$ to ensure that $\tilde{f}(p) = \infty \in V$. The composition $\varphi = w \circ \tilde{f} \circ z^{-1} : z(U \cap \tilde{f}^{-1}(V)) \to \mathbb{C}$ is continuous at $z(p)$ as it is a composition of continuous functions. Hence we can calculate the limit

$$\lim_{x \to z(p)} |\varphi(x)| = \lim_{x \to z(p)} |w(\tilde{f}(z^{-1}(x)))| = |w(\tilde{f}(z^{-1}(z(p))))| = 0.$$

In particular we see that $z(p)$ is a removable singularity of $\varphi$ as the above limit exists and is finite. Hence we can extend $\varphi$ holomorphically at $z(p)$ and thus $\tilde{f} : M \to \mathbb{P}^1\mathbb{C}$ is a holomorphic map between Riemann surfaces.
Conversely, if we have a holomorphic map $f : M \to \mathbb{P}^1\mathbb{C}$ between Riemann surfaces, it is either the constant map with value $\infty \in \mathbb{P}^1\mathbb{C}$, or the preimage $f^{-1}(\infty)$ is discrete (as we have seen in Remark 2.11). In the latter case, let $\tilde{M}$ denote $M \setminus f^{-1}(\infty)$. The restriction $f|_{\tilde{M}} : \tilde{M} \to \mathbb{C}$ is a meromorphic function. In conclusion, there is a correspondence between meromorphic functions on a Riemann surface $M$ and holomorphic maps between the Riemann surfaces $M$ and $\mathbb{P}^1\mathbb{C}$ that are not constant with value $\infty$.

**Example 2.22.** We follow the Example 5.3. in [V96] to characterize the meromorphic functions $\mathcal{M}(\mathbb{P}^1\mathbb{C})$ on the Riemann sphere. As we have seen in Paragraph 2.21 above, we can view the meromorphic functions as holomorphic maps from the Riemann sphere to itself not constant at $\infty$.
Let $P$ and $Q$ be two polynomials with complex coefficients, and assume that $Q \neq 0$. Consider the map

$$f : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C},\ x \mapsto \frac{P(x)}{Q(x)}$$

with the usual conventions for the points $\infty \in \mathbb{P}^1\mathbb{C}$. As a quotient of two holomorphic functions where the denominator is not 0 everywhere, the map $f$ is a meromorphic function on $\mathbb{P}^1\mathbb{C}$ because $\mathcal{M}(\mathbb{P}^1\mathbb{C})$ is a field containing the holomorphic functions.
Conversely, let $f$ be any meromorphic function on $\mathbb{P}^1\mathbb{C}$. We show $f$ is a rational function, that is, $f$ is the quotient of two polynomials as above. Since $\mathbb{P}^1\mathbb{C}$ is compact, by Remark 2.11 the function $f$ has finitely many poles, denote them

by $p_1, ..., p_s$. We consider first the case that all of the poles are in $\mathbb{C}$. For each pole $p_i$ we can write $f$ as a Laurent extension $f = \sum_{j=-k_i}^{\infty} c_{i,j}(z-p_i)^j$ around $p_i$. Consider the meromorphic functions

$$f_i : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C},\ f_i(z) = \sum_{j=-k_i}^{-1} c_{i,j}(z-p_i)^j$$

for $i = 1, ..., s$. Note that for each $i = 1, ..., s$, the function $f - f_i$ has no pole at $p_i$, and $f_i$ has only one pole, which is $p_i$. Hence by taking the difference $g = f - \sum_{i=1}^{s} f_i$, we get a meromorphic function without poles. In particular, $g$ is a holomorphic function on $\mathbb{P}^1\mathbb{C}$, so it must be constant by Remark 2.14. It follows that $f = g + \sum_{i=1}^{s} f_i$ is rational.
In the other case, $f$ has a pole at $\infty$ so the function $1/f$ will satisfy the assumption from the first case that all poles are in $\mathbb{C}$. This implies $1/f$ will be rational and thus $f$ as well. In conclusion, we have shown that the meromorphic function on the Riemann sphere $\mathbb{P}^1\mathbb{C}$ are the rational functions.

## 2.2 Degree and ramification

This section further studies holomorphic maps between Riemann surfaces. We will introduce the degree and ramification index of a holomorphic map. These concepts help us to better understand how holomorphic maps behave. We begin with a lemma which is the inverse function theorem for holomorphic functions. With that result, we will then be able to see how holomorphic maps behave locally. At the end, we also look at the monodromy of a holomorphic map. For further reference, we refer again to the textbooks mentioned at the beginning of Chapter 2.1.

**Lemma 2.23.** *Let $f : U \to \mathbb{C}$ be a holomorphic function on an open set $U \subset \mathbb{C}$ and $p \in U$ a point such that the derivative $f'(p) \neq 0$. There exists an open neighbourhood $U' \subset U$ of $p$, such that the restriction $f|_{U'} : U' \to f(U')$ is a holomorphic bijection with holomorphic inverse, that is, the restriction is biholomorphic.*

*Proof.* We will use the inverse function theorem for real functions. Thus, as in the proof of Proposition 2.5, we view the complex function $f$ as a real function in two variables. For any $z = x + iy \in U$, write $f(z) = u(x, y) + iv(x, y)$ where $u$ and $v$ are real-valued functions on the open set $U$ viewed as a subset of $\mathbb{R}^2$. With this notation, the function $f$ can be viewed as

$$f : U \subset \mathbb{R}^2 \to \mathbb{R}^2,\ (x, y) \mapsto f(x, y) = (u(x, y), v(x, y)).$$

The determinant of the Jacobian $J_f(x_0, y_0)$ of $f$ at the point $p = x_0 + iy_0$ is

$$\det(J_f(x_0, y_0) = |f'(p)|^2 > 0,$$

as we can see from the calculation in the proof of Proposition 2.5 and the Cauchy-Riemann equations. Hence we can apply the inverse function theorem for real functions in two variables, which gives us a neighbourhood $U' \subset U$ of the point $p$ such that the restriction $f|_{U'} : U' \to f(U')$ is bijective and the inverse $g = (f|_{U'})^{-1}$ is a smooth map, because $f$ viewed as a real map is smooth. It is only left to show that $g$ is holomorphic. From the inverse function theorem in the real case we know that the Jacobian of $g$ at a point $q' = f(p') \in f(U')$ is given by the inverse matrix

$$J_g(q) = (J_f(p))^{-1}. \tag{1}$$

Since $f$ is holomorphic, its partial derivatives satisfy the Cauchy-Riemann equations. With the relation in (1), it also follows that the imaginary and real part of the function $g$ satisfy the Cauchy-Riemann equations and thus $g$ is holomorphic. □

**Proposition 2.24.** *Let $f : M \to N$ be a non-constant holomorphic map between Riemann surfaces. For every point $p \in M$, there exists a unique integer $r \geq 1$ such that there exists a chart $(U, z)$ of $M$ with $p \in U$ and a chart $(V, w)$ of $N$ around $q = f(p)$. These charts further satisfy that the composition $w \circ f \circ z^{-1}$ is equal to the map $x \mapsto x^r$ on its domain of definition.*

*Proof.* Let $(\tilde{U}, \tilde{z})$ and $(V, w)$ be arbitrary charts of $M$ and $N$ around the points $p \in M$ and $q = f(p) \in N$ respectively. By composing the charts with a translation, which is holomorphic, we can assume without loss of generality that the charts satisfy $\tilde{z}(p) = 0$ and $w(q) = 0$. The composition

$$\psi = w \circ f \circ (\tilde{z})^{-1} : \tilde{z}(\tilde{U} \cap f^{-1}(V)) \to w(V)$$

is a holomorphic function satisfying $\psi(0) = 0$. Thus for some $r \in \mathbb{Z}_{\geq 1}$ and $a_i \in \mathbb{C}$ for $i \geq r$ where $a_r \neq 0$, the Taylor expansion of $\psi$ around 0 is

$$\psi(x) = \sum_{k=r}^{\infty} a_k x^k = x^r \sum_{k=0}^{\infty} a_{k+r} x^k.$$

The holomorphic function on a neighbourhood of 0 given by

$$h(x) = \sum_{k=0}^{\infty} a_{k+r} x^k$$

satisfies $h(0) = a_r \neq 0$ and by continuity we can find a disc $D \subset \mathbb{C}$ around 0 such that $h(x) \neq 0$ for every point $x \in D$. Thus there exists a branch of $\log(h)$ in $D$, let us denote it by $\varphi$. The function

$$g : D \to \mathbb{C},\ x \mapsto x \cdot \exp(\varphi(x)/r)$$

is holomorphic on $D$ and satisfies

$$g(x)^r = x^r \cdot \exp(\varphi(x)) = x^r \cdot h(x) = \psi(x)$$

for every $x \in D$. Note that the derivative $g'(0) = 1 \neq 0$, hence by Lemma 2.23 we can restrict $g$ to another neighbourhood $\tilde{D} \subset D$ of 0 on which $g$ is a holomorphic bijection with holomorphic inverse.
Finally, the composition

$$z = g \circ \tilde{z} : \tilde{U} \cap \tilde{z}^{-1}(\tilde{D}) \to \mathbb{C}$$

is a homeomorphism and thus with $U = \tilde{U} \cap \tilde{z}^{-1}(\tilde{D})$, the pair $(U, z)$ is a chart on $M$. The transition functions will be holomorphic because $g$ is biholomorphic. The composition $w \circ f \circ z^{-1}$ satisfies the desired property because for every $x$ in its domain of definition:

$$(w \circ f \circ z^{-1})(x) = (w \circ f \circ (\tilde{z})^{-1} \circ g^{-1})(x) = \psi(g^{-1}(x)) = (g(g^{-1}(x)))^r = x^r.$$

To conclude the proof, we show uniqueness of the integer $r$. Note that if the composition $w \circ f \circ z^{-1}$ as above is given by mapping $x \mapsto x^r$, for points $q'$ near $q = f(p)$ but $q' \neq q$, there are $r$ preimages $f^{-1}(q')$ because the composition has $r$ preimages and the maps $w$ and $z$ are bijections. This shows that the number $r$ is independent of the choice of charts and determined only by $f$, hence the uniqueness follows. □

The proposition above describes the local behaviour of any holomorphic, non-constant map between Riemann surfaces. It states in other words that any such map can be viewed locally as the map that sends $x \mapsto x^r$ for some $r \geq 1$. We say that the map $x \mapsto x^r$ represents the holomorphic map $f$ locally.

**Definition 2.25.** In the setting from Proposition 2.24, the number $r$ is called the *ramification index* or *multiplicity* of $f$ at the point $p \in M$. We denote it by $r_f(p)$. We call a point $p \in M$ that satisfies $r_f(p) > 1$ a *ramification point*. The image of a ramification point will be called a *branch point*.

**2.26.** From the proof of Proposition 2.24 above, we can further see how to calculate the ramification index of the holomorphic map $f : M \to N$ at a point $p \in M$ given any two charts $(U, z)$ of $M$ with $p \in U$, and $(V, w)$ of $N$ with $q = f(p) \in V$. Consider the composition $h = w \circ f \circ z^{-1}$, and the translation $T_a : \mathbb{C} \to \mathbb{C}, x \mapsto x + a$. The maps

$$\begin{aligned} \tilde{z} &= z - z(p) = T_{-z(p)} \circ z \\ \tilde{w} &= w - w(q) = T_{-w(q)} \circ w \end{aligned}$$

are also charts and achieve that $\tilde{z}(p) = 0$ and $\tilde{w}(q) = 0$, which is the generalization we first made in the proof. The ramification index $r_f(p)$ appears in the proof as the order of the zero at the point 0 of the holomorphic map $\tilde{w} \circ f \circ \tilde{z}^{-1}$. Since

$$(\tilde{w} \circ f \circ \tilde{z}^{-1}) = (T_{-w(q)} \circ w) \circ f \circ (T_{-z(p)} \circ z)^{-1} = T_{-w(q)} \circ h \circ T_{z(p)},$$

we can see that the the Taylor expansion of $h$ around $z(p)$ is of the form

$$h(x) = h(z(p)) + \sum_{k=n}^{\infty} a_k (x - z(p))^k.$$

Hence the ramification index $r_f(p) = r$ also equals the lowest exponent greater than or equal to 1 in the Taylor expansion of $h$ around $z(p)$. Note that if we denote the order of the zero $z(p)$ of the derivative $h'$ by $m$, then the ramification index also equals $r_f(p) = m + 1$.

**Proposition 2.27.** *Let $f : M \to N$ be a non-constant holomorphic map between Riemann surfaces. The set $R \subset M$ of ramification points of $f$ is a discrete subset of $M$.*

*Proof.* By the previous Paragraph 2.26, the ramification points of $f$ correspond to the zeros of the derivatives of the local representations of $f$. Since derivatives of holomorphic functions are holomorphic themselves, their zeros are discrete by a result from complex analysis. It follows that the set $R$ is discrete in $M$. □

**Definition 2.28.** The *total branching number* $b_f$ of a non-constant holomorphic map $f : M \to N$ between compact Riemann surfaces is the sum

$$b_f = \sum_{p \in M} (r_f(p) - 1).$$

**2.29.** Proposition 2.27 ensures that the total branching number is well defined. Indeed, the ramification points $R \subset M$ of $f$ form a discrete subset in $M$ and the set $R$ is closed, since by Proposition 2.24 about the local representation of $f$, it follows that every point $r \in M \setminus R$ has an open neighbourhood which does not contain any ramification points. Thus the set of ramification points is a closed, discrete subset of a compact space $M$ and thus must be finite. In particular, there are only finitely many non-zero summands in the sum defining the total branching number.

**Example 2.30.** Consider the non-constant holomorphic map $p : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C}$ given by the polynomial $P = \sum_{i=0}^{n} a_i X^i \in \mathbb{C}[X]$ with $a_n \neq 0$ as in Example 2.9. We further use the chart $(U_2, z_2)$ of $\mathbb{P}^1\mathbb{C}$ around $\infty$ as in Example 2.3. Note that this chart satisfies $p(\infty) = \infty \in U_2$. We can write the composition

$$z_2 \circ p \circ z_2^{-1} : z_2(\mathbb{P}^1\mathbb{C} \setminus (\{0\} \cup f^{-1}(0))) \to \mathbb{C},$$
$$x \mapsto x^n \left( \frac{1}{a_0 x^n + a_1 x^{n-1} + ... + a_n} \right)$$

where the second factor does not vanish at 0. We have found a factorisation as in the proof of Proposition 2.24 and hence the ramification index $r_p(\infty)$ will be $n$, which is also the degree of the polynomial $P$.

In the following, we define the degree of a holomorphic map. The degree will be the number of preimages at any point in the range of the holomorphic map, but to guarantee that this number is well-defined, we have to count the preimages with their multiplicities, as the following lemma shows.

**Lemma 2.31.** *Let $f : M \to N$ be a non-constant holomorphic map between compact Riemann surfaces. The number*

$$d(q) = \sum_{p \in f^{-1}(q)} r_f(p)$$

*is independent of $q \in N$.*

**Remark 2.32.** Before we prove the lemma, note that the sum is finite for every point $q \in N$. Indeed the set $f^{-1}(q)$ is closed because $f$ is continuous and by Remark 2.11 it is discrete. Since $M$ is compact, it follows that $f^{-1}(q)$ is finite.

*Proof.* In the case where $f : \mathbb{C} \to \mathbb{C}$ is given by $x \mapsto x^r$ for some $r \in \mathbb{Z}_{\geq 1}$, the statement holds because for $q = 0 \in \mathbb{C}$ the only preimage is 0 and the ramification index $r_f(0) = r$, hence $d(q) = r$. For every other point $q' \in \mathbb{C} \setminus \{0\}$ the preimage consists of the $r$ distinct $r$th roots of $q'$. At each of these roots the ramification index is 1, so $d(q') = r$ and the statement holds for this choice of $f$.
For a general non-constant holomorphic map $f : M \to N$ we show that the map $d : N \to \mathbb{N}$, $q \mapsto d(q)$ is locally constant. Since the Riemann surface $N$ is connected, it follows that $d$ is globally constant on $N$.
Let $q \in N$ be an arbitrary point and the preimage $f^{-1}(q) = \{p_1, ..., p_m\}$. By Proposition 2.24, for every $i = 1, ..., m$ there exist local coordinates $(U_i, z_i)$ of $M$ such that $p_i \in U_i$ and $z_i(p_i) = 0$, and a local coordinate $(V, w)$ of $N$ around $q$ with $w(q) = 0$, and such that for each $i = 1, ..., m$ the map $f$ is locally represented by $x \mapsto x^{r_i}$. To get the open neighbourhood $V$, just take the intersection of the finitely many open neighbourhoods of $q$ for each $i$ from Proposition 2.24.
We can further restrict $V$ to an open neighbourhood $V' \subset V$ of $q$ such that for every $y \in V'$, the preimages

$$f^{-1}(y) \subset U = \bigcup_{i=1,...,m} U_i.$$

Indeed, assume that this is not the case. Then there are points arbitrarily close to $q \in N$ with preimages not contained in $U$. Using these preimages we can find a sequence $\{x_j\}_{j\geq 1}$ of points in $M \setminus U$ such that the sequence of the images $\{f(x_j)\}_{j\geq 1}$ in $N$ converges to $q$. By assumption $M$ is compact, so there is a converging subsequence $\{x_{j_n}\}_{n\geq 1}$ with limit point $x \in M$. The function $f$ is continuous, so

$$f(x) = \lim_{n\to\infty} f(x_{j_n}) = q.$$

This implies $x = p_{i'}$ for some $i' \in \{1, .., m\}$ which leads to a contradiction as for every $n \geq 1: \ x_{j_n} \notin U_{i'}$. Using this, it follows from the first case that $d(q)$ is locally constant in $V'$. $\square$

**Definition 2.33.** In the setting of Proposition 2.31, we call the number $n = d(q)$ for any $q \in N$ the *degree* of $f$ and denote it by $\deg(f)$. For a constant holomorphic map $f : M \to N$ between Riemann surfaces, we set $\deg(f) = 0$.

**Remark 2.34.** Note that by choosing a point $q \in N$ which is not a branch point, the degree of $f$ simplifies to

$$\deg(f) = \sum_{p\in f^{-1}(q)} r_f(p) = \sum_{p\in f^{-1}(q)} 1 = |f^{-1}(q)|,$$

the number of preimages of $q$ under the map $f$.

**Example 2.35.** Let $P$ be a polynomial of degree $n$, we compute the degree of the induced holomorphic map $p : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C}$ as described in the Example 2.9. The preimage $p^{-1}(\infty) = \{\infty\}$ consists only of one point, thus the degree of $p$ is the ramification index at that point. We have seen in Example 2.30 that the ramification index of $p$ at $\infty$ is $r_p(\infty) = n$, hence the degree $\deg(p) = n = \deg(P)$.

**Example 2.36.** In this example we calculate the degree of a rational function

$$f : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C}, x \mapsto f(x) = \frac{P(x)}{Q(x)}$$

where $P, Q \in \mathbb{C}[X]$ are two coprime polynomials with degrees $\deg(P) = n, \deg(P) = m$ and $Q \neq 0$.
We first consider the case $n > m$ and calculate the degree of $f$ by counting the multiplicities of the preimages $f^{-1}(0)$. Clearly the zeros $\{a_1, ..., a_r\}$ of the polynomial $P$ are zeros of $f$. Since $n > m$ we have $f(\infty) = \infty$, so there are no other zeros of $f$ except for $\{a_1, ..., a_r\}$. The polynomials $P$ and $Q$ are coprime, so the multiplicity $r_f(a_i)$ for any $1 \leq i \leq r$ equals the multiplicity $\alpha_i$ of the zero $a_i$ from the polynomial $P$. This holds because we can factorize the Taylor expansion of $f$ around the point $a_i$ as

$$f(x) = (x - a_i)^{\alpha_i} h(x)$$

where $h$ is a function that satisfies $h(a_i) \neq 0$. Thus $\deg(f) = \sum_{i=1}^{r} \alpha_i = n$ by the fundamental theorem of algebra.
In the case where $m > n$ we calculate the degree by looking at the preimages of $\infty$. Since in this case $f(\infty) = 0$, the only preimages $f^{-1}(\infty)$ are the zeros $\{b_1, ..., b_s\}$ of $Q$. To calculate the multiplicities at those points, we need to consider the composition $z_2 \circ f$ where $z_2 : \mathbb{P}^1\mathbb{C} \setminus \{0\} \to \mathbb{C}, x \mapsto 1/x$ is the chart of $\mathbb{P}^1\mathbb{C}$, as we know from Example 2.3. The composition satisfies

$$(z_2 \circ f)(x) = \frac{Q(x)}{P(x)}$$

for any $x$ in its domain of definition. The zeros $\{b_1, ..., b_s\}$ of the polynomial $Q$ are zeros of $z_2 \circ f$ and we conclude, as in the first case, that the multiplicities $r_f(b_i)$ equal the respective multiplicities of the zero $b_i$ of $Q$ and hence $\deg(f) = m$.
For $m = n$, we can either look at zeros of $f$ or the preimage $f^{-1}(\infty)$ to deduce that $\deg(f) = m = n$ similar to the cases above.
Considering all the cases we conclude that

$$\deg(f) = \max\{\deg(P), \deg(Q)\}.$$

We will now see how holomorphic maps between compact Riemann surfaces are related to covering maps. Recall the following definition.

**Definition 2.37.** A *covering map* between two topological spaces $X$ and $\tilde{X}$ is a continuous map $p : \tilde{X} \to X$ with the following property. For every point $x_0 \in X$ there exists an open neighbourhood $U$ of $x_0$ such that the preimage $p^{-1}(U) = \bigsqcup_{j=1}^{n} V_j$ is a disjoint union of open subsets $V_j \subset \tilde{X}$. Furthermore, for every $j = 1, .., n$ the restriction $p|_{V_j} : V_j \to U$ is a homeomorphism.

**2.38.** Let $f : M \to N$ be a non-constant holomorphic map between compact Riemann surfaces with degree $\deg(f) = n$. Denote by $B \subset N$ the set of all branch points of $f$, which is finite as a consequence of 2.29. The restriction

$$\tilde{f} = f|_{M \setminus f^{-1}(B)} : M \setminus f^{-1}(B) \to N \setminus B$$

has ramification index 1 at every point $p \in M$ and hence we can find open neighbourhoods $U$ of $p$ and $V$ of $f(p)$ such that $f|_U$ is a homeomorphism, that is, $\tilde{f}$ is a local homeomorphism. Furthermore, since no point $q \in N$ is a branch point, there are precisely $n$ preimages of $q$. Since the degree $n$ is finite and $N \setminus B$ is connected, it follows that $\tilde{f}$ is an $n$-sheeted covering map.
Due to this, we call holomorphic maps satisfying the same properties as $f$ also $n$-sheeted *holomorphic* or *branched covering maps*. If such a map has no branch points, we call it an unbranched map and it is an $n$-sheeted covering in this case. To emphasize that we mean a covering map as in Definition 2.37 above, we also say a topological covering map. See also [F95, Proposition 19.3.].

**2.39.** In this paragraph, we will introduce a notion that will appear again later in Chapter 4, as a reference, we used [M95, Chapter III.4.]. To begin, let $p : \tilde{X} \to X$ be a topological $n$-sheeted covering map, thus the fiber $p^{-1}(q)$ of any point $x_0 \in X$ consists of $n$ points and we label them by $x_1, ..., x_n$. A loop $\gamma \in \pi_1(X, x_0)$ of the fundamental group of $X$ can be lifted to $n$ paths $\tilde{\gamma}_i : [0,1] \to \tilde{X}$ for $1 \leq i \leq n$ where $\tilde{\gamma}_i(0) = x_i$. The endpoints of the lifted paths $\tilde{\gamma}_i$ will also be in the fiber of $x_0$, as $\gamma$ is a loop and thus we can write $\tilde{\gamma}_i(1) = x_{\sigma(i)}$ for some function $\sigma : \{1, ..., n\} \to \{1, ..., n\}$. Recall the following result about topological coverings that can be found for example in [B93, Corollary 3.5.].

**Theorem 2.40** (Monodromy theorem)**.** *Let $p : \tilde{X} \to X$ be a covering map and $\tilde{\alpha}, \tilde{\beta}$ paths in $\tilde{X}$ such that their images $\alpha = p \circ \tilde{\alpha}$ and $\beta = p \circ \tilde{\beta}$ are homotopic with fixed endpoints in $X$. The starting points $\tilde{\alpha}(0) = \tilde{\beta}(0)$ are equal if and only if the endpoints $\tilde{\alpha}(1) = \tilde{\beta}(1)$ are equal.*

It follows from this theorem that the function $\sigma$ we defined above is in fact a permutation and only depends on the homotopy class of $\gamma$. Thus the covering $p$ induces a group homomorphism $\rho : \pi_1(X, x_0) \to S_n$, where $S_n$ denotes the symmetric group of the set of $n$ elements.

**Definition 2.41.** The *monodromy representation* of an $n$-sheeted covering map $p : \tilde{X} \to X$ is the above group homomorphism $\rho : \pi_1(X, x_0) \to S_n$.

**Proposition 2.42.** *The action of the set $\{1, ..., n\}$ given by the monodromy representation $\rho : \pi_1(X, x_0) \to S_n$ of an $n$-sheeted covering map $p : \tilde{X} \to X$ is transitive if $\tilde{X}$ is path-connected.*

*Proof.* Let $\tilde{X}$ be path-connected and the preimage $p^{-1}(x_0) = \{x_1, ..., x_n\}$, choose arbitrary indices $1 \leq i, j \leq n$. Since $\tilde{X}$ is path-connected, there exists a path $\tilde{\gamma} : [0,1] \to \tilde{X}$ with $\tilde{\gamma}(0) = x_i$ and $\gamma(\tilde{1}) = x_j$. The composition $\gamma = p \circ \tilde{\gamma}$ is a loop in $X$, since both $x_i$ and $x_j$ are in the fiber of $x_0$. By the construction of the monodromy representation $\rho$ the permutation $\rho([\gamma])$ sends $i$ to $j$. Because $i$ and $j$ were arbitrary, it follows that the action is transitive. □

With the preceding paragraph 2.38 we are able to extend the definition of monodromy representations to holomorphic maps between Riemann surfaces.

**Definition 2.43.** The *monodromy representation* of a holomorphic map $f : M \to N$ between compact Riemann surfaces with branch points $B \subset N$ is the monodromy representation $\rho : \pi_1(N \setminus B, x_0) \to S_n$ of the covering map $f|_{M \setminus f^{-1}(B)}$.

**Remark 2.44.** Since we consider compact Riemann surfaces, the set of branch points of $f$ is finite and the degree of $f$ is also finite, thus we only remove finitely many points from $M$ if we consider $M \setminus f^{-1}(B)$. It follows that $M \setminus f^{-1}(B)$ is still path-connected and by Proposition 2.42 the action given by the monodromy representation of $f$ is transitive.

Since we know the local behaviour of holomorphic maps, we can describe its monodromy representation according to the following proposition. A proof can be found in [M95, Lemma 4.6.].

**Proposition 2.45.** *Let $f : M \to N$ be a holomorphic map between compact Riemann surfaces and $b \in N$ a branch point with $k$ preimages $x_1, ..., x_k$. The cycle structure of the permutation $\rho([\gamma])$ given by the image of a small loop $\gamma$ around $b$ under the monodromy representation $\rho$ of $f$ is $(r_f(x_1), ..., r_f(x_k))$.*

**Remark 2.46.** To define a small loop $\gamma$ around a branch point $b \in N$, we consider an open neighbourhood $W \subset N$ around $b$ such that the preimage of $W$ under the map $f$ is a disjoint union of open neighbourhoods around the preimages of $b$ under $f$. A small loop around $b$ is a loop starting at the base point $x_0 \in N \setminus B$, then following a path $\alpha$ from $x_0$ to any point $x \in W \setminus b$, going around $b$ in $W$ with winding number one, and finally returning to $x_0$ by following the path $\alpha$ connecting $x_0$ to $x$ in reverse direction.

**2.47.** In the following, we will quickly give a second way to define the degree of a holomorphic map between Riemann surfaces that coincides with the degree given in Definition 2.33. Let $f : M \to N$ be a holomorphic map between Riemann surfaces. We consider the fields of meromorphic functions $\mathcal{M}(M)$ and $\mathcal{M}(N)$ on $M$ and $N$ respectively. The map $f$ induces a map

$$\begin{aligned} f^* : \mathcal{M}(N) &\to \mathcal{M}(M), \\ \varphi &\mapsto f^*(\varphi) = \varphi \circ f \end{aligned}$$

given by the pull-back. The pull-back $f^*$ is a field homomorphism and hence is injective. This allows us to view $\mathcal{M}(N)$ as a subfield of $\mathcal{M}(M)$ and thus to study the degree of the field extension $\mathcal{M}(M)/\mathcal{M}(N)$. The following proposition shows the relation to the degree of the map $f$, although we will not prove the proposition here. It is proven in a more general way in [H92, Proposition 7.16.].

**Proposition 2.48.** *The degree of a holomorphic map $f : M \to N$ between Riemann surfaces equals the degree of the field extension $\mathcal{M}(M)/\mathcal{M}(N)$.*

## 2.3 Triangulations

The main result we will give in this section is that any compact Riemann surface can be triangulated. Hence, roughly speaking, we can build any Riemann surface by glueing triangles together in a suitable way. First, we will start with the necessary definitions. We will call a connected topological 2-manifold a topological surface.

**Definition 2.49.** A *triangulation* of a compact topological surface $S$ is a decomposition

$$S = \bigcup_{i=1}^{n} T_i$$

into finitely many closed subsets $T_i$ such that each of these is homeomorphic to a triangle $T$ in $\mathbb{R}^2$ via the homeomorphism $\varphi_i : T \to T_i$. We call the image $T_i \subset S$ a triangle for every $1 \leq i \leq n$, and the images under $\varphi$ of the vertices and edges of the triangle $T$ will be called vertices and edges as well. The triangles $T_i$ in the decomposition of $S$ have to be disjoint, or intersect only at a single vertex, or along an entire single edge.

**Theorem 2.50.** *Every compact Riemann surface can be triangulated.*

The above theorem follows from the more general result that every compact topological surface can be triangulated, which we will not prove here. A proof of the more general result can be found in [T92], where the Jordan Schönflies Theorem is being used.
We will deduce the specific case for Riemann surfaces in Chapter 3 as a corollary from the Riemann existence theorem. We will study two properties about triangulations next, which we will need for the proof of the Riemann-Hurwitz formula in Chapter 3 and the above mentioned corollary.

**Lemma 2.51.** *Let $S$ be a compact topological surface and $P \subset S$ be a finite set of points. There exists a triangulation on $S$ such that every point $p \in P$ is a vertex.*

*Proof.* By Theorem 2.50 there exists a triangulation of $S$. A point $p \in P$ can either be a vertex, lie on an edge, or in the interior of a triangle in the triangulation of $S$. In the first case where $p$ is already a vertex, there is nothing to do.
In the second case, $p$ lies on an edge $e$ of the triangle $T \subset S$. Denote the triangle in the plane $\mathbb{R}^2$ homeomorphic to $T$ by $T'$, and the edge and point in $T'$ corresponding to $e$ and $p$ by $e'$ and $p'$. Adding an edge $d'$ connecting $p'$ with the vertex opposite to the edge $e'$ divides $T'$ into two new triangles such that $p'$ is a vertex. See also Figure 1 for clarification. If we add the image of the new edge $d'$ under the homeomorphism between the triangles $T$ and $T'$ to the triangulation, the point $p$ becomes a vertex.
The last case is when $p$ lies in the interior of a triangle $T \subset S$ homeomorphic to a triangle $T' \subset \mathbb{R}^2$ with the point $p'$ corresponding to $p$. We can add three

new edges connecting $p'$ to each of the vertices of $T'$. This divides $T'$ into three new triangles such that $p'$ is a vertex. Again, by using the homeomorphism between $T$ and $T'$, we add these three new edges to the triangulation on $S$ to achieve that $p$ is a vertex. □

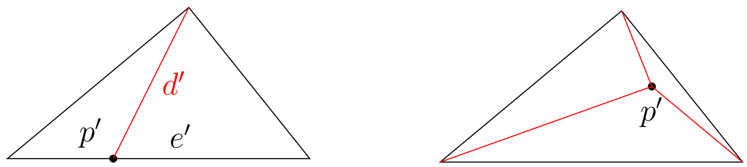
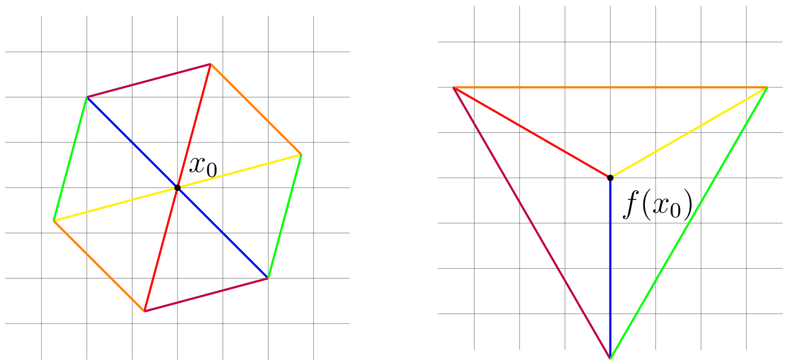



Figure 1: Illustration of the two non-trivial cases of the proof for Lemma 2.51; the red edges are the new ones we add to the existing triangulation

**2.52.** As a next property, we will see how a non-constant holomorphic map $f : M \to N$ between compact Riemann surfaces induces a triangulation on $M$ if we are given a triangulation on $N$. First, note that since the Riemann surfaces are compact, the set of branch points $B \subset N$ is finite as we can see from Paragraph 2.29. Thus with the above Lemma 2.51, we can ensure that every branch point is a vertex of the triangulation. The idea is that the preimage of the triangulation on $N$ under the map $f$ defines a triangulation on $M$. To motivate that this indeed defines a triangulation, we first recall again that the restriction $f : M \setminus f^{-1}(B) \to N \setminus B$ is a covering map of degree $n = \deg(f)$. Hence away from branch points, each triangle in $N$ lifts to $n$ homeomorphic triangles in $M$. We also need to verify that at branch points the preimage is again a triangulation. Doing this only requires local knowledge about $f$, so we can use the Proposition 2.24 to reduce to the case where the map $f$ is given by $z \mapsto z^r$, which we can better understand. In this case the only branch point is 0 and the ramification point is 0. In the Figure 2 below, one can see schematically what happens in the case for $r = 2$. In particular, we see that the preimage of the triangulation gives a triangulation again.

## 2.4 Cohomology and genus

In this section we will briefly discuss cohomology and homology of a Riemann surface in order to define its genus. In particular, we will discuss simplicial, singular, and de Rham cohomology. Since we are interested in Riemann surfaces, we will only introduce the homology and cohomology for surfaces and not in more generality. We use the word surface to denote a real manifold of dimension 2. We will also not provide proofs of the results we use, some references to fill in the missing details and proofs are [H01, Chapters 2-3] for simplicial and singular homology and cohomology, for de Rham cohomology we refer to [BT82, Chapter 1] and [L13, Chapters 17-18] . At the end of this




(a) preimage (b) triangulation on $N$

Figure 2: Schematic picture of the preimage of a triangulation under a holomorphic map $f$, where $x_0$ is a ramification point of $f$ with ramification index $r_f(x_0) = 2$

section, we also introduce the Euler characteristic of a surface and relate it to the genus.

**2.53.** To introduce simplicial and singuar homology, we first need to define $n$-simplices. An $n$-simplex is an $n$-dimensional analog of the triangle for every $n \in \mathbb{Z}_{\geq 0}$, although we will only be concerned with 0-, 1-, and 2-simplices here. The standard $n$-simplex is defined as

$$\Delta^n = \{(t_0, ..., t_n) \in \mathbb{R}^{n+1} | \sum_{i=0}^{n} t_i = 1 \text{ and } \forall\, 0 \leq i \leq n: \ t_i \geq 0\}.$$

Note that the vertices of the standard simplex are the standard unit vectors in $\mathbb{R}^{n+1}$, see also Figure 3.

A general $n$-simplex is defined by points $v_0, ..., v_n$ in a Euclidean space $\mathbb{R}^m$ for some $m \geq n$ such that the differences $v_1 - v_0, ..., v_n - v_0$ are linearly independent. We denote the $n$-simplex defined by these points with $[v_0, ..., v_n]$ and it is defined as the convex hull of $v_0, ..., v_n$, that is, the smallest convex set in $\mathbb{R}^m$ that contains $v_0, ..., v_n$. Note that $v_0, ..., v_n$ are the vertices of this $n$-simplex.

For any $n$-simplex $[v_0, ..., v_n]$ there is a homeomorphism

$$\Delta^n \to [v_0, ..., v_n], \ (t_0, ..., t_n) \mapsto \sum_{i=0}^{n} t_i v_i$$

that preserves the ordering of the vertices, so we will identify the standard $n$-simplex with $[v_0, ..., v_n]$ using this homeomorphism.

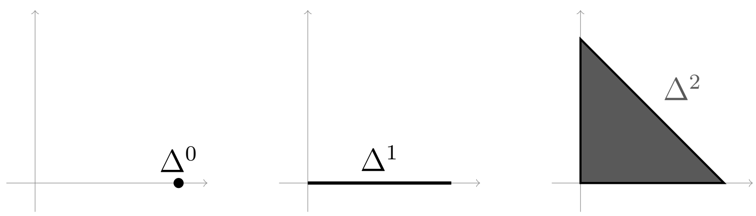



Figure 3: Illustration of the 0-, 1-, and 2-dimensional standard simplices

**2.54.** We first look at simplicial homology, which is defined for triangulated topological surfaces. Such a surface $X$ comes with maps $\sigma_\alpha : \Delta^2 \to X$ with image $\sigma_\alpha(\Delta^2) = T_\alpha$ for every triangle $T_\alpha$ in $X$, where $\alpha$ is in some finite index set $I_2$. Analogous collections of maps obtained from restricting every $\sigma_\alpha$ are $\{\sigma_\beta : \Delta^1 \to X\}_{\beta \in I_1}$ and $\{\sigma_\gamma : \Delta^0 \to X\}_{\gamma \in I_0}$ for the edges and vertices of $X$. Define the free abelian group $\Delta_n(X)$ for $n = 0, 1, 2$ as the free abelian group generated by the open $n$-simplices $\sigma_\alpha(\mathrm{Int}(\Delta^n))$. An element of $\Delta_n(X)$ is called $n$-chain and is a finite formal sum that can be represented as $\sum_{\alpha \in I_n} z_\alpha \sigma_\alpha$, where the coefficient $z_\alpha$ are integers.
The maps

$$\partial_2 : \Delta_2(X) \to \Delta_1(X),\ \partial_2([v_0, v_1, v_2]) = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$$

and

$$\partial_1 : \Delta_1(X) \to \Delta_0(X),\ \partial_1([v_0, v_1]) = v_1 - v_0$$

are called differentials or boundary operators. A calculation shows that $\partial_2 \circ \partial_1 = 0$, thus we have a chain complex

$$0 \xrightarrow{\partial_3} \Delta_2(X) \xrightarrow{\partial_2} \Delta_1(X) \xrightarrow{\partial_1} \Delta_0(X) \xrightarrow{\partial_0} 0 \tag{2}$$

of free abelian groups, where $\partial_3 = 0$ and $\partial_2 = 0$. The homology of this chain complex, which is defined to be the abelian group

$$H_n^\Delta(X) = \ker(\partial_n)/\mathrm{im}(\partial_{n+1})$$

for $n = 0, 1, 2$, is called the $n$th simplicial homology group of $X$. We set $H_n^\Delta(X) = 0$ for every $n \neq 0, 1, 2$.

**2.55.** As a next homology theory we want to give the example of singular homology. Singular homology generalizes simplicial homology and is defined for any topological space $X$. Since the space $X$ does not come with maps from the standard simplex to $X$, we take all the continuous maps $\sigma : \Delta^n \to X$ for any $n \in \mathbb{Z}_{\geq 0}$ into consideration. Such a map $\sigma$ is called a singular $n$-simplex and the free abelian group generated by all $n$-simplices is denoted by $S_n(X)$. The elements of $S_n(X)$ are called (singular) $n$-chains, one can define $n$-chains

for any $n \in \mathbb{Z}_{\geq 0}$, but for our purpose the space $X$ is a topological surface and thus it will be enough to consider 0-,1-, and 2-chains and set all the other $S_n(X)$ to 0.
The idea is again to build a chain complex of free abelian groups with the groups $S_n(X)$, so we need to define the boundary operators. Similarly to the boundary of the simplicial chain complex (2), we set

$$\begin{aligned}\partial_2 : S_2(X) \to S_1(X),\ \sigma \mapsto &\partial_2(\sigma) = \sigma|_{[e_1,e_2]} - \sigma|_{[e_0,e_2]} + \sigma|_{[e_0,e_1]}\\ \partial_1 : S_1(X) \to S_0(X),\ \alpha \mapsto &\partial_1(\alpha) = \alpha(1) - \alpha(0).\end{aligned}$$

With these boundary operators, where again $\partial_n = 0$ for every $n \neq 1, 2$, one can check that

$$0 \xrightarrow{\partial_3} S_2(X) \xrightarrow{\partial_2} S_1(X) \xrightarrow{\partial_1} S_0(X) \xrightarrow{\partial_0} 0 \tag{3}$$

defines a chain complex of free abelian groups and thus we can take the homology groups

$$H_n(X) = \ker(\partial_n)/\mathrm{im}(\partial_{n+1})$$

for any $n \in \mathbb{N}$ which we will call singular homology of $X$.
Note that for a triangulated surface $X$, the groups $S_n(X)$ are much larger than the groups $\Delta_n(X)$, and thus it is not immediately clear that the quotients $H_n(X)$ will be for example finitely generated. But there are things easier to see using singular homology instead of simplicial homology, such as singular homology being isomorphic for homeomorphic spaces.
Both simplicial and singular homology are defined on triangulated topological surfaces and one can show that they are in fact isomorphic to each other. A proof of this can be found in [H01, Theorem 2.27.]. The canonical isomorphism $H_n^\Delta(X) \to H_n(X)$ for any triangulated topological surface $X$ and for any $n \in \mathbb{Z}$ is induced by the inclusion $\Delta_n(X) \to S_n(X)$.

**2.56.** In this paragraph, we will define singular and simplicial cohomology, to do that we fix an abelian group $G$. Singular cohomology $H^n(X; G)$ with coefficient group $G$ is defined by taking the cohomology of the dual chain complex of (3). To be more explicit, the dual chain complex of (3) is

$$0 \xleftarrow{\partial_3^*} \hom(S_2(X), G) \xleftarrow{\partial_2^*} \hom(S_1(X), G) \xleftarrow{\partial_1^*} \hom(S_0(X), G) \xleftarrow{\partial_0^*} 0.$$

The differentials $\partial_n^*$ are the dual maps of $\partial_n$, so any $f \in \hom(S_{n-1}(X), G)$ is mapped to $\partial_n^*(f) = f \circ \partial_n$. This gives a chain complex of free abelian groups and its cohomology groups will be defined as

$$H^n(X; G) = \ker(\partial_{n+1}^*)/\mathrm{im}(\partial_n^*).$$

In an analogous way one defines simplicial cohomology $H_\Delta^n(X, G)$ by using the

dual of the chain complex (2). The isomorphism between the singular and simplicial homology for any triangulated topological surface $X$ also induces isomorphisms $H^n_\Delta(X;G) \cong H^n(X;G)$ between singular and simplicial cohomology for any $n \in \mathbb{Z}$ and any abelian group $G$.
For a space where the homology groups are finitely generated, the homology groups determine the cohomology groups and vice versa. They are related via the universal coefficients theorem and in the case where the coefficients group $G = \mathbb{R}$, the relation

$$H^n(X;\mathbb{R}) \cong \hom_G(H_n(X), \mathbb{R})$$

holds. In particular, it follows that $H^n(X;\mathbb{R})$ is also a real vector space.

**2.57.** The last cohomology theory we want to introduce is the de Rham cohomology. We will first show how the de Rham cohomology is defined on $\mathbb{R}^2$. Denote the coordinates of $\mathbb{R}^2$ by $x$ and $y$. We introduce the formal symbols $dx$ and $dy$ that satisfy the relations

$$\begin{aligned} (dx)^2 = (dy)^2 &= 0 \\ dx\,dy &= -dy\,dx \end{aligned} \tag{4}$$

and define $\Omega^*$ to be the algebra over $\mathbb{R}$ generated by $dx$ and $dy$. With the set $C^\infty(\mathbb{R}^2)$ of smooth functions on $\mathbb{R}^2$, define the algebra

$$\Omega^*(\mathbb{R}^2) = C^\infty(\mathbb{R}^2) \otimes_{\mathbb{R}} \Omega^*$$

which consists of the so called $C^\infty$ differential forms on $\mathbb{R}^2$. The algebra

$$\Omega^*(\mathbb{R}^2) = \bigoplus_{q=0}^{2} \Omega^q(\mathbb{R}^2)$$

is naturally graded and we call the elements of $\Omega^q(\mathbb{R}^2)$ the $C^\infty$ $q$-forms (or just $q$-forms for short) on $\mathbb{R}^2$, which we will now describe further.
The 0-forms on $\mathbb{R}^2$ are precisely the smooth functions on $\mathbb{R}^2$. A 1-form is a formal sum $f\,dx + g\,dy$ where $f$ and $g$ are smooth functions on $\mathbb{R}^2$, and the expression $f\,dx\,dy$ is a 2-form for every smooth function $f$ on $\mathbb{R}^2$.
The set of the $q$-forms are vector spaces over $\mathbb{R}$ and they will form a chain complex of vector spaces. The corresponding differential operator

$$d : \Omega^q(\mathbb{R}^2) \to \Omega^{q+1}(\mathbb{R}^2)$$

is defined for any $f \in \Omega^0(\mathbb{R}^2)$ as

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

and for any $\omega = f\,dx + g\,dy \in \Omega^1(\mathbb{R}^2)$ we define

$$d\omega = df\,dx + dg\,dy = \left(\frac{\partial g}{\partial y} - \frac{\partial f}{\partial x}\right) dx\,dy.$$

Note that the second equality comes from the definition of $df$ and $dg$ and using the relations between $dx$ and $dy$ stated in (4).
The composition $d^2$ of the differential with itself is 0, by Schwarz's Theorem on the symmetry of the second partial derivatives. Thus we get a chain complex of vector spaces

$$0 \to \Omega^0(\mathbb{R}^2) \xrightarrow{d} \Omega^1(\mathbb{R}^2) \xrightarrow{d} \Omega^2(\mathbb{R}^2) \to 0$$

called the de Rham complex on $\mathbb{R}^2$. The cohomology $H^q_{DR}(\mathbb{R}^2)$ of this complex is called de Rham cohomology of $\mathbb{R}^2$, it is a vector space over $\mathbb{R}$. By taking $C^\infty$ functions on an open set $U \in \mathbb{R}^2$, one can analogously define the de Rham cohomology $H^q_{DR}(U)$ of $U$.
Since we are interested in surfaces, we will briefly explain the idea of how the definition of de Rham cohomology can be extended to surfaces. On a surface $S$ with atlas $\{(U_\alpha, z_\alpha)\}_{\alpha\in I}$, a function $f$ on $U_\alpha$ is smooth if the composition $f \circ z_\alpha^{-1}$ is smooth. This allows us to define a differential $q$-form on any $U_\alpha$ in the same way as before. A collection $\{\omega_\alpha\}_{\alpha\in I}$ where each $\omega_\alpha$ is a $q$-form on $U_\alpha$ defines a $q$-form on the whole surface $S$ if it satisfies some additional compatibility conditions on the intersections $U_\alpha \cup U_\beta$. We will not use these compatibility conditions explicitly, they can be found in [BT82, Chapter 1.2].

**2.58.** An additional structure on $\Omega^*(\mathbb{R}^2)$ is the wedge product

$$\wedge : \Omega^p(\mathbb{R}^2) \times \Omega^q(\mathbb{R}^2) \to \Omega^{p+q}(\mathbb{R}^2).$$

We will only need the wedge product of two 1-forms $\omega = f\,dx + g\,dy$ and $\tau = u\,dx + v\,dy$. It is defined as

$$\omega \wedge \tau = fv\,dx\,dy + gu\,dy\,dx = (fv - gu)dx\,dy.$$

Note that the wedge product is graded commutative, that is, $\omega \wedge \tau = -\tau \wedge \omega$.

**2.59.** De Rham cohomology and singular cohomology are both defined on surfaces. The de Rham theorem states that in this case the de Rham cohomology is isomorphic to the singular cohomology with real coefficients. One can find a proof of this theorem for example in Chapter 18 of [L13], we will only describe the isomorphism between the two cohomology theories here.
The idea is to integrate a differential $q$-form $\omega$ on a surface $S$ over a $q$-simplex of $S$. To ensure that everything is well defined, one has to take a smooth

$q$-simplex which is a map $\sigma : \Delta^q \to S$ that has a smooth extension to a neighborhood of every point $p \in \Delta^q$. This allows us to pull-back the $q$-form $\omega$ by $\sigma$ to get a $q$-form $\sigma^*(\omega) = \omega \circ \sigma$ on $\Delta^q$ which we can integrate, so

$$\int_\sigma \omega = \int_{\Delta^q} \sigma^*(\omega).$$

For any homology class $a \in H_q(S)$, one can find a smooth $q$-cycle $\tilde{\sigma}$ that represents $a$. Thus we can define the map

$$\Psi : H^q_{DR}(S) \to H^q(S;\mathbb{R}) \cong \hom(H_q(S);\mathbb{R})$$

by assigning to any de Rham cohomology class $\alpha \in H^q_{DR}(S)$ represented by the $q$-form $\omega$ the homomorphism that sends a singular homology class $a \in H_q(M)$ represented by the smooth singular chain $\tilde{\sigma}$ to

$$\Psi(\alpha)(a) = \int_{\tilde{\sigma}} \omega.$$

Stokes' theorem ensures that $\Psi$ is well defined and one can show that $\Psi$ is an isomorphism of real vector spaces.

With the concept of cohomology, we define the genus of a Riemann surface.

**Definition 2.60.** The *genus* of a compact Riemann surface $M$ is the number

$$g(M) = \frac{\dim(H^1_{DR}(M))}{2}.$$

**Remark 2.61.** We have seen above in 2.56 and 2.59 that the de Rham, singular, and simplicial cohomology with real coefficients are isomorphic. Thus, in particular, they all have the same dimension and one can use any of these cohomology theories to define the genus.

**2.62.** To see that the genus is a non-negative integer, it is convenient to study the dimension of the first de Rham cohomology $H^1_{DR}(M)$ of a compact Riemann surface $M$. With the wedge product from 2.58 we define the map

$$\begin{aligned} \Phi : H^1_{DR}(M) \times H^1_{DR}(M) &\to \mathbb{R} \\ (\omega, \tau) &\mapsto \int_M \omega \wedge \tau. \end{aligned}$$

One can show that this is a non-degenerate, antisymmetric bilinear form on $H^1_{DR}(M)$ and thus the dimension of $H^1_{DR}(M)$ must be even. A proof can be found in [D11, Chapter 5.2].

One can think about the genus in a less formal way as the number of "holes" of a Riemann surface. The classification of topological surfaces states that any compact oriented topological surface is homeomorphic to a connected sum of $n \geq 1$ tori and thus it has $n$ holes, or it is homeomorphic to the sphere which has zero holes. One can compute the cohomology of the sum of $n$ connected tori to see that the dimension of the first cohomology group will be $2n$, and the first cohomology group of the sphere is 0, so this interpretation agrees with our Definition 2.60 of the genus.

**Definition 2.63.** The *Euler characteristic* $\chi(X)$ of a compact Riemann surface $M$ is the sum

$$\chi(M) = \sum_{i=0}^{2} (-1)^i \operatorname{rank}(H_i(X)).$$

**2.64.** By calculating the homology of a compact Riemann surface $M$, one finds the relation

$$g(M) = \frac{1}{2}(2 - \chi(M)) \tag{5}$$

between the genus and the Euler characteristic of $M$.
The Riemann surface $M$ is in particular a manifold with real dimension 2, so for $i \neq 0, 1, 2$ the homology groups $H_i(M)$ are zero and the only thing left is to calculate the ranks of $H_0(M), H_1(M)$, and $H_2(M)$.
A Riemann surface is connected by definition, hence the homology group in degree 0 is $H_0(M) \cong \mathbb{Z}$ and so $\operatorname{rank} H_0(M) = 1$. For every orientable compact surface, the second homology group is $H_2(M) \cong \mathbb{Z}$. The first homology group can be expressed in terms of the genus, since the isomorphism $H^1(M; \mathbb{R}) \cong \hom(H_1(M), \mathbb{R})$ given by the universal coefficients theorem implies that $\operatorname{rank}(H_1(M)) = \dim(H^1(M; \mathbb{R})) = 2g(M)$, where we use the definition of the genus.
Thus the Euler characteristic is $\chi(M) = 1 - 2g(M) + 1$ and by rearranging the equation we get the relation (5).

**2.65.** In this paragraph, we will see how to calculate the Euler characteristic of a compact topological surface $S$ using a triangulation. By Theorem 2.50, there exists a triangulation of $S$, denote by $f, e, v$ the number of faces, edges, and vertices respectively. Since we have seen that singular homology is isomorphic to simplicial homology, we can use the rank of the simplicial homology groups to calculate the Euler characteristic. We claim that

$$\chi(S) = \sum_{i=0}^{2} (-1)^i \operatorname{rank}(H_i^{\Delta}(X)) = v - e + f. \tag{6}$$

This claim is a result about chain complexes of finitely generated abelian groups applied to the simplicial chain complex

$$0 \xrightarrow{\partial_3} \Delta_2(X) \xrightarrow{\partial_2} \Delta_1(X) \xrightarrow{\partial_1} \Delta_0(X) \xrightarrow{\partial_0} 0.$$

We follow the proof given in [H01, Theorem 2.44.]. The definitions of a chain complex and its homology group ensure that the sequences

$$0 \to \ker(\partial_n) \to \Delta_n(X) \to \operatorname{im}(\partial_n) \to 0$$
$$0 \to \operatorname{im}(\partial_{n+1}) \to \ker(\partial_n) \to H_n^{\Delta}(X) \to 0$$

are exact for $n = 0, 1, 2$. These short exact sequences of finitely generated abelian groups imply the following relations between the ranks of the groups

$$\operatorname{rank}(\ker(\partial_n)) + \operatorname{rank}(\operatorname{im}(\partial_n)) = \operatorname{rank}(\Delta_n(X))$$
$$\operatorname{rank}(\operatorname{im}(\partial_{n+1})) + \operatorname{rank}(H_n^{\Delta}(X)) = \operatorname{rank}(\ker(\partial_n)).$$

Substituting the expression of $\operatorname{rank}(\ker(\partial_n))$ into the first equation and rearranging the terms we get

$$\operatorname{rank}(H_n^{\Delta}) = \operatorname{rank}(\Delta_n(X)) - \operatorname{rank}(\operatorname{im}(\partial_n)) - \operatorname{rank}(\operatorname{im}(\partial_{n+1})). \tag{7}$$

Note that

$$\operatorname{rank}(\Delta_n(X)) = \begin{cases} v, & \text{for } n = 0 \\ e, & \text{for } n = 1 \\ f, & \text{for } n = 2 \end{cases} \tag{8}$$

since $\Delta_n(X)$ is the free abelian group generated by the vertices, edges, or faces depending on $n$. Using the expressions from (7) and (8) to calculate the Euler characteristic yields

$$\chi(X) = v - e + f.$$

**Example 2.66.** With the relations (5) and (6) we can compute the genera of simple topological surfaces for which we can find a triangulation. The Riemann sphere is homeomorphic to the tetrahedron, this homeomorphism defines a triangulation of the sphere with 4 vertices, 6 edges, and 4 faces. Hence the Euler characteristic of the sphere is $\chi(\mathbb{P}^1\mathbb{C}) = 4 - 6 + 4 = 2$ and the genus is $g(\mathbb{P}^1\mathbb{C}) = \frac{1}{2}(2 - 2) = 0$.
A triangulation of the 2-dimensional torus $\mathbb{T}$ as indicated in Figure 4 gives $\chi(\mathbb{T}) = 4 - 12 + 8 = 0$ and thus the genus $g(\mathbb{T}) = \frac{1}{2}(2 - 0) = 1$. Note that we used the usual way to construct the 2-dimensional torus by identifying opposite edges of the unit square as indicated in the figure.

# 3 Fundamental theorems about Riemann surfaces

In this chapter, we will see some fundamental results about Riemann surfaces. We state the Riemann-Roch theorem, the uniformization theorem and the

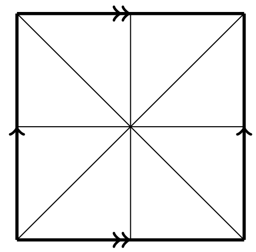


Figure 4: Triangulation of the torus used in Example 2.66 to compute the genus of the torus

Riemann existence theorem, but we will not prove them here as it is not in the scope of this thesis. The last theorem of this section is the Riemann-Hurwitz formula, for which we will give a proof and some concrete examples to demonstrate how to work with the formula.

## 3.1 The Riemann-Roch theorem

The Riemann-Roch theorem describes the dimension of the vector space of meromorphic functions satisfying some additional conditions on a compact Riemann surface. The proof of the theorem and further details we could not cover here can be found for example in [J97, Chapter 5.4] and [S14, Chapter 7]. To state the Riemann-Roch theorem we need to introduce some new definitions and notation. Throughout this whole section, let $M$ be a compact Riemann surface.

**Definition 3.1.** A *divisor* on $M$ is a finite formal sum

$$D = \sum_{i=1}^{n} s_i P_i$$

where $P_i \in M$ are points and the coefficients $s_i \in \mathbb{Z}$. Equivalently, a divisor is an element of the free abelian group $\mathrm{Div}(M)$ generated by the points of $M$.

**Definition 3.2.** A divisor $D = \sum_{i=1}^{n} s_i P_i$ on $M$ is *effective* if $s_i \geq 0$ for all $i = 1, ..., n$. If the difference $D - D'$ of two divisors on $M$ is effective, we write $D \geq D'$.

For every meromorphic function $f \not\equiv 0$ on $M$ one can define a divisor $(f)$ associated to $f$ in the following way. Define the order $\mathrm{ord}_P(f)$ of a point $P \in M$ to be equal to $k$ if $P \in M$ is a zero of $f$ of order $k$, and $\mathrm{ord}_P(f) = -k$ if $P$ is a pole of $f$ of order $k$. In every other case we set $\mathrm{ord}_P(f) = 0$. Define the divisor $(f)$ of the meromorphic function $f$ by

$$(f) = \sum_{P \in f^{-1}\{0,\infty\}} (\mathrm{ord}_P(f))P.$$

Since $M$ is compact, there are only finitely many poles and zeros of $f$, so the sum defining $(f)$ is finite. Divisors of this form have a special name.

**Definition 3.3.** A divisor $D$ is called *principal divisor* if there exists a meromorphic function $g \neq 0$ on $M$ such that $D = (g)$.

It is clear from the definitions that $f$ is meromorphic if and only if $(f) \geq 0$, as this is precisely the case where $f$ does not have any poles.
Similarly to the $C^\infty$ differential forms we introduced in 2.57, one can define meromorphic differentials. A meromorphic 1-form $\eta$ is locally given by $\eta = f\,dz$ where $(U, z)$ is a local coordinate of $M$ and $f$ is a meromorphic function on $U$. Thus we can define $\mathrm{ord}_P(\eta) = \mathrm{ord}_P(f)$ for points $P \in U$, this definition is independent of the chosen local chart around $P$. Analogously as for divisors of meromorphic functions, we define the divisor

$$(\eta) = \sum_{P \in M} (\mathrm{ord}_P(\eta))P \tag{9}$$

of a 1-form $\eta \not\equiv 0$. Again, only finitely many summands in (9) are non-zero because of the compactness of $M$, and we also give this form of divisor a name.

**Definition 3.4.** A *canonical divisor* $K$ is the divisor $(\eta)$ of a meromorphic 1-form $\eta \not\equiv 0$ on $M$.

**Definition 3.5.** The divisors $D_1$ and $D_2$ on $M$ are *linearly equivalent* if their difference is a principal divisor.

Two canonical divisors $K = (\eta)$ and $K' = (\eta')$ are linearly equivalent, since $\eta/\eta' = f$ is a meromorphic function and thus $K - K' = (f)$.

**3.6.** We define the complex vector space

$$\mathcal{L}(D) = \{g \in \mathcal{M}(M) \mid g \equiv 0 \text{ or } D + (g) \geq 0\}$$

and denote its dimension $\dim_{\mathbb{C}} \mathcal{L}(D)$ by $h^0(D)$.
If the divisors $D$ and $D'$ are linearly equivalent, the vector spaces $\mathcal{L}(D)$ and $\mathcal{L}(D')$ are isomorphic. Indeed, there exists a non-zero meromorphic function $g$ on $M$ such that $D - D' = (g)$. The homomorphism

$$\begin{aligned} \mathcal{L}(D) &\to \mathcal{L}(D') \\ f &\mapsto gf \end{aligned}$$

is an isomorphism of vector spaces, hence in particular $h^0(D) = h^0(D')$.

**Definition 3.7.** The *degree* $\deg(D)$ of a divisor $D = \sum_{i=1}^n s_i P_i$ on $M$ is the integer

$$\deg(D) = \sum_{i=1}^n s_i.$$

Note that the degree defines a map $\deg : \mathrm{Div}(M) \to \mathbb{Z}$, which is a homomorphism of groups.
For a meromorphic function $f$ on $M$, we have seen in 2.31 that the number of poles of $f$ counted with multiplicities equals the number of zeros of $f$ counted with multiplicities, so the principal divisor $D = (f)$ has degree $\deg(D) = 0$. This further implies that linearly equivalent divisors have the same degree. With these notions we can state the Riemann-Roch theorem.

**Theorem 3.8** (Riemann-Roch)**.** *Let $D$ be a divisor of a compact Riemann surface $M$ with genus $g(M)$. For any canonical divisor $K$, it holds*

$$h^0(D) = 1 - g(M) + \deg(D) + h^0(K - D).$$

**Remark 3.9.** Since any two canonical divisors are linearly equivalent, by Paragraph 3.6 the dimension $h^0(K - D)$ is independent of the choice of the canonical divisor $K$ in the theorem.

The Riemann-Roch theorem gives the dimension of the space $\mathcal{L}(D)$ for a divisor $D = \sum_{i=1}^{n} s_i P_i$. One can describe the vector space $\mathcal{L}(D)$ as the space of all meromorphic functions on $M$ such that their poles are points $P_i \in M$ where $s_i > 0$ and the order of the pole is at most $s_i$. Furthermore, they have zeros at all points $P_i$ where $s_i < 0$ and the order of the zero is greater than or equal than $-s_i$.

## 3.2 The uniformization theorem

The uniformization theorem classifies the simply connected Riemann surfaces. It is easy to state, so we do not need to introduce new concepts. A proof of the theorem as well as more details and historic background can be found in the book [SG16], a briefer resource is [WMLI92, Chapters 2, 7].

**Theorem 3.10** (Uniformization)**.** *Every simply connected Riemann surface is holomorphically isomorphic to the complex plane, the Riemann sphere, or the open unit disc in $\mathbb{C}$ which is isomorphic to the upper half plane $\mathfrak{h} = \{z \in \mathbb{C} \mid Im(z) > 0\}$.*

The uniformization theorem can also be stated in terms of universal coverings of Riemann surfaces. The universal cover is simply connected by definition and every Riemann surface has a universal cover, because as a connected manifold it is path-connected, locally path-connected, and semi-locally simply connected. Thus the uniformization theorem can be reformulated to state that every Riemann surface has a universal cover that is holomorphically isomorphic to the Riemann sphere, the complex plane, or the open unit disc in $\mathbb{C}$. This allows us to classify Riemann surfaces according to their universal coverings.

**Definition 3.11.** We say that a Riemann surface is

- *elliptic* if and only if its universal cover is isomorphic to the Riemann sphere $\mathbb{P}^1\mathbb{C}$;
- *parabolic* if and only if its universal cover is isomorphic to the complex plane $\mathbb{C}$;
- *hyperbolic* if and only if its universal cover is isomorphic to the open unit disc in $\mathbb{C}$.

The Riemann sphere is the only elliptic Riemann surface up to holomorphic isomorphism.
We have seen elliptic curves in Example 2.16 as a quotient of the complex plane, thus elliptic curves are parabolic Riemann surfaces. The only other elliptic Riemann surfaces are holomorphically isomorphic either to $\mathbb{C}$ or the punctured plane $\mathbb{C}^*$.
Hyperbolic Riemann surfaces can be expressed as a quotient $\mathfrak{h}/\Gamma$ for some subgroup $\Gamma < \mathrm{PSL}(2, \mathbb{R})$ satisfying certain additional conditions.
If we consider only compact Riemann surfaces, we can relate the classification in Definition 3.11 to the genus of the Riemann surface. A compact Riemann surface of genus 0 is elliptic, a compact Riemann surface of genus 1 is parabolic, and if the genus of a compact Riemann surface is $\geq 2$, then the Riemann surface is hyperbolic.

## 3.3 The Riemann existence theorem

In this rather short section we state two equivalent versions of the Riemann existence theorem. Afterwards, we show how to deduce the existence of a triangulation on a compact Riemann surface from this result.

**Theorem 3.12** (Riemann existence)**.** *There exists a non-constant meromorphic function on every compact Riemann surface.*

**Theorem 3.13** (Riemann existence)**.** *For any distinct points $p_0, ..., p_n$ of a compact Riemann surface $M$ and complex numbers $c_1, ..., c_n \in \mathbb{C}$ there exists a meromorphic function $f$ on $M$ with $f(p_i) = c_i$ for every $i = 1, ..., n$.*

One can use the classification of Riemann surfaces into hyperbolic, elliptic, and parabolic, which we saw in the preceding section to prove the Riemann existence theorem. Such a proof can be found in [FK92, Chapter IV.3]. For an alternative approach using the theory of Hilbert spaces and cohomology we refer to the proof in [V96, Chapter 6].
To conclude this section, we prove Theorem 2.50 about the existence of a triangulation as a corollary of the Riemann existence theorem.

*Proof of Theorem 2.50.* For the Riemann sphere, the existence of a triangulation follows from the homeomorphism between the tetrahedron and the sphere. For the general case, let $M$ be a compact Riemann surface. By the Riemann existence theorem, there exists a non-constant meromorphic function on $M$. We have seen that this corresponds to a non-constant holomorphic map $f : M \to \mathbb{P}^1\mathbb{C}$. As in the proof of Lemma 2.51, we can refine the triangulation of the Riemann sphere such that each branch point of $f$ is a vertex. Note again that because of compactness there are only finitely many branch points. The inverse image of this refined triangulation on the sphere $\mathbb{P}^1\mathbb{C}$ under the map $f$ gives a triangulation on $M$ as we have explained in 2.52. $\square$

## 3.4 The Riemann-Hurwitz formula

After we have introduced all the needed concepts, we state and prove the Riemann-Hurwitz formula in this section. The formula links the degree and total branching number of a holomorphic map between compact Riemann surfaces with the genera of the Riemann surfaces. At the end we will show two examples on how to calculate with the Riemann-Hurwitz formula.

**Theorem 3.14** (Riemann-Hurwitz formula)**.** *Let $f : M \to N$ be a non-constant holomorphic map between compact Riemann surfaces. The genera $g(N)$ and $g(M)$ of the Riemann surfaces $N$ and $M$ respectively, the total branching number $b_f$ of $f$ and the degree $d = deg(f)$ are related by*

$$g(M) - 1 = d \cdot (g(N) - 1) + \frac{b_f}{2}.$$

*Proof.* We will follow the proof of the Riemann-Hurwitz formula given in [M95, Chapter II.4]. By the results of Section 2.3 about triangulations, there exists a triangulation $T$ on the Riemann surface $N$ such that every branch point of $f$ is a vertex of $T$. Denote the number of vertices, edges, and triangles of $T$ by $v$, $e$, and $t$ respectively. Furthermore, the preimage of this triangulation defines a triangulation $T'$ on the Riemann surface $M$, say with $v'$ vertices, $e'$ edges and $t'$ triangles. Because the edges and triangles of the triangulation on $N$ consist only of unbranched points of the map $f$, they each have $d$ distinct preimages under $f$, that is,

$$e' = d \cdot e, \quad t' = d \cdot t.$$

Let $q \in N$ be an arbitrary vertex of the triangulation $T$. Since $q$ might be a branch point of $f$, it will not have $d$ preimages in general. We can express the number of preimages of $q$ using the definition of the degree $d$ of $f$ by

$$|f^{-1}(q)| = d - \sum_{p \in f^{-1}(q)} r_f(p) + |f^{-1}(q)| = d - \sum_{p \in f^{-1}(q)} (r_f(p) - 1).$$

Using this result, the number $v'$ of vertices of the triangulation on $M$ can be computed as

$$\begin{aligned}
v' &= \sum_{\text{vertex } q \text{ of } T} |f^{-1}(q)| \\
&= \sum_{\text{vertex } q \text{ of } T} \left( d - \sum_{p \in f^{-1}(q)} (r_f(p) - 1) \right) \\
&= d \cdot v - \sum_{\text{vertex } q \text{ of } T} \left( \sum_{p \in f^{-1}(q)} (r_f(p) - 1) \right) \\
&= d \cdot v - \sum_{\text{vertex } p \text{ of } T'} (r_f(p) - 1) \\
&= d \cdot v - \sum_{p \in M} (r_f(p) - 1).
\end{aligned}$$

The last equation holds because all ramification points of $f$ are vertices of the triangulation $T'$ on $M$. Hence every point $p \in M$ that is not a vertex has ramification index $r_f(p) = 1$, and thus the additional summands in the last step are all equal to 0. Note also that the sum in the last equation is precisely the total branching number $b_f = \sum_{p \in M}(r_f(p) - 1)$.
With the expressions for $v'$, $e'$, and $f'$ we have derived above, we calculate the Euler characteristic of $M$ as in Paragraph 2.65

$$\begin{aligned}
\chi(M) &= v' - e' + t' \\
&= dv - b_f - de + dt \\
&= d(v - e + t) - b_f \\
&= d\chi(N) - b_f.
\end{aligned}$$

To conclude, we use the relation between the Euler characteristic and the genus of a surface derived in 2.64, the Riemann-Hurwitz formula

$$g(M) - 1 = d \cdot (g(N) - 1) + \frac{b_f}{2}$$

follows. □

An immediate conclusion from the Riemann-Hurwitz formula is that the total branching number is always even, as the degree of a holomorphic map and the genus of a Riemann surface are integers.
Since the total branching number is greater than or equal to 0, one can deduce that if there exists a non-constant holomorphic map between compact Riemann surfaces $M$ and $N$, then the genera satisfy $g(M) \geq g(N)$. In particular $g(M) = 0$ implies $g(N) = 0$.

**Example 3.15.** To see how concrete calculations with the Riemann-Hurwitz formula work, we will verify the Riemann-Hurwitz formula for a non-constant holomorphic map between the Riemann sphere and itself. As we have already seen, these are the rational maps. We will study the map $f : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C}$ corresponding to the meromorphic function on $\mathbb{P}^1\mathbb{C}$ that maps

$$z \mapsto \frac{4z^3(z-1)}{(2z-1)^2}$$

for any $z \in \mathbb{P}^1\mathbb{C}$.
From 2.36 we know the degree of $f$ is $\deg(f) = 4$ and the genus of the Riemann sphere is 0, as we have seen in Example 2.66. To find all ramification points and their ramification indices, we use the approach from 2.26. In that paragraph we used the local representation of $f$, which is the composition

$$\varphi = w \circ f \circ z^{-1} : z(U \cap f^{-1}(V)) \to w(V) \tag{10}$$

for charts $(U, z), (V, w)$ of $\mathbb{P}^1\mathbb{C}$ to calculate the ramification points and their ramification indices. If we take the chart $(\mathbb{C}, id)$ for both charts in (10), the composition is just $f$ restricted to $\mathbb{C} \setminus \{1/2\}$. The derivative of $f$ can be calculated as

$$f'(z) = \frac{4z^2(4z^2-6z+3)}{(2z-1)^3} = \frac{z^2(-4iz+\sqrt{3}+3i)(4iz+\sqrt{3}-3i)}{(2z-1)^3}$$

for any $z \in \mathbb{C} \setminus \{1/2\}$. We saw that the zeros of $f'$ correspond to the ramification points of $f$, and the order of the zero of $f'$ is one less than the ramification index at that point. Because we use the identity function as a chart here, the zeros of $f'$ equal the ramification points. Hence we get the following ramification points and indices:

- $z_0 = 0$ with $r_f(z_0) = 3$,
- $z_1 = \frac{1}{4}(3 - \sqrt{3}i)$ with $r_f(z_1) = 2$,
- $z_2 = \frac{1}{4}(3 + \sqrt{3}i)$ with $r_f(z_2) = 2$.

By using the charts $(\mathbb{C}, id)$, we only find ramifications points in $\mathbb{C} \subset \mathbb{P}^1\mathbb{C}$ with their image also in $\mathbb{C} \subset \mathbb{P}^1\mathbb{C}$.
Thus it is only left to check whether $\infty$ is a branch point, and if so which are the corresponding ramification points and indices. The preimage of $\infty$ is

$$f^{-1}(\infty) = \{\infty, \frac{1}{2}\}.$$

To find the ramification index of $\frac{1}{2}$, we take the charts $(U, z) = (\mathbb{C}, id)$ and $(V, w) = (\mathbb{P}^1\mathbb{C} \setminus \{0\}, z \mapsto 1/z)$. The derivative of the composition in (10) is

$$\varphi'(z) = -\frac{(2z-1)(4z^2-6z+3)}{4z^4(z-1)^2}.$$

From this derivative we see that the point $z_4 = \frac{1}{2}$ is a ramification point with $r_f(z_4) = 2$. It follows that $z_5 = \infty$ is also a ramification point with ramification index $r_f(z_5) = 2$, because the degree of $f$ is 4 and thus the relation $r_f(z_4) + r_f(z_5) = 4$ must hold.
Finally, we can calculate the total branching number $b_f = 2+1+1+1+1 = 6$. Plugging everything into the Riemann-Hurwitz formula gives

$$0 - 1 = 4(0-1) + \frac{6}{2},$$

which evaluates to $-1$ on both sides.

**Example 3.16.** In the next example we use the Riemann-Hurwitz formula to calculate the genus of a Riemann surface that we do not know already. The Riemann surface we study is given by

$$M = \{[x, y, z] \in \mathbb{PC}^2 \,|\, y^2 z = \prod_{k=1}^{3}(x - a_k z)\}$$

for some mutually distinct points $a_1, a_2, a_3 \in \mathbb{C}$. Since $M$ is a closed subset of the compact space $\mathbb{PC}^2$, it is compact itself. We need to check that $M$ is indeed a Riemann surface. To do this, note that $M$ is the union of the two subsets $M_1, M_2 \subset M$ where

$$M_1 = \{[x, y, z] \in M | y \neq 0\}$$
$$M_2 = \{[x, y, z] \in M | z \neq 0\}.$$

Since the point $[1, 0, 0] \notin M$, it follows that $M_1$ and $M_2$ do indeed cover $M$. If we can show that both $M_1$ and $M_2$ have no singularities, that is, the Jacobians of the functions defining $M_1$ and $M_2$ have full rank at every point, then we can conclude that $M$ is a Riemann surface by the implicit function theorem.
The set $M_1$ consists of points where $y \neq 0$, and since we are in the projective space we can assume without loss of generality that $y = 1$, and can write $M_1$ as the set

$$\{M_1 = (x, z) \in \mathbb{C}^2 \,|\, z - \prod_{k=1}^{3}(x - a_k z) = 0\}.$$

The Jacobian of the function $F : \mathbb{C}^2 \to \mathbb{C}, (x, z) \mapsto z - \prod_{k=1}^{3}(x - a_k z)$ at a point $p = (x, z) \in M_1$ is

$$D_p F = (-\sum_{i=1}^{3}\prod_{k \neq i}(x - a_k z), 1 + \sum_{i=1}^{3} a_i \prod_{k \neq i}(x - a_k z)).$$

Notice that $z = 0$ implies $x = 0$ for a point $(x, z) \in M_1$. In this case the Jacobian is $D_pF = (0, 1)$, so it has full rank. If $z \neq 0$, we can go over to study $M_2$. Analogously as for $M_1$, we have

$$M_2 = \{(x, y) \in \mathbb{C}^2 \mid y^2 - \prod_{k=1}^{2g+1} (x - a_k) = 0\}.$$

The function $G : \mathbb{C}^2 \to \mathbb{C}, (x, y) \mapsto y^2 - \prod_{k=1}^{2g+1}(x - a_k)$ has at any point $p = (x, y) \in M_2$ the Jacobian

$$D_pG = (-\sum_{i=1}^{3} \prod_{k \neq i} (x - a_k), 2y).$$

If $y \neq 0$, then $D_pF$ certainly has rank 1. If $y = 0$, we need to check that $-\sum_{i=1}^{3} \prod_{k \neq i}(x - a_k) \neq 0$. For points $(x, y) \in M$ where $y = 0$, it follows

$$\prod_{k=1}^{3} (x - a_k) = 0$$

and thus $x = a_{k'}$ for some $k' = 1, 2, 3$. All zeros $a_k$ of $\prod_{k=1}^{3}(x - a_k)$ are simple, thus in particular the derivative at $a_{k'}$ is not zero. More concretely,

$$\sum_{i=1}^{3} \prod_{k \neq i} (a_{k'} - a_k) = \prod_{k \neq k'} (a_{k'} - a_k) \neq 0$$

since the $a_k$ are mutually distinct, and we conclude that $M$ is a Riemann surface.

This example can be found in more generality in [GG12, Chapter 1] where the reader can also find an atlas of $M$ and thus a reasoning that the map

$$f : M \to \mathbb{P}^1\mathbb{C},\ [x, y, z] \mapsto \begin{cases} \frac{x}{z}, & \text{for } z \neq 0 \\ \infty, & \text{for } z = 0 \end{cases}$$

is holomorphic. To find the degree of $f$, note that every point on the sphere except $a_1, a_2, a_3, \infty$ has two preimages, so the degree of $f$ is $\deg(f) = 2$. The ramification points are thus $a_1, a_2, a_3, \infty$ and each of their ramification indices must be 1, which is the only possibility, since a ramification point has ramification index less than the degree of the map and greater to or equal than 1. We know the genus of the Riemann sphere is 0, so we have calculated everything in the Riemann-Hurwitz formula except $g(M)$, which we can thus compute as

$$g(M) = 1 + 2 \cdot (0 - 1) + \frac{4}{2} = 1.$$

# 4 Davenport-Zannier polynomials and dessins d'enfants

We are now able to study the polynomial Catalan conjecture in this last section, for which we mostly follow Chapters $1-2$ from the book [APZ20]. Recall that the polynomial Catalan conjecture concerns coprime polynomials $A, B \in \mathbb{C}[X]$ with degrees $\deg(A) = 2k,\ \deg(B) = 3k$ for $k > 0$. The goal is to find the minimal degree of the difference $A^3 - B^2$. The reason we included this problem here is because there is a link to meromorphic functions on the Riemann sphere, and the Riemann-Hurwitz formula will be an important tool to solve the problem.
We will introduce a generalized version of the above problem and show results towards proving the polynomial Catalan conjecture, which we will do at the end of the chapter.

**Definition 4.1.** A tuple $\alpha = (\alpha_1, .., \alpha_p)$ where $\alpha_i \in \mathbb{Z}_{\geq 0}$ for every $i = 1, ..., p$ is a *partition* of the number $n \in \mathbb{Z}_{\geq 0}$ if $\sum_{i=1}^{p} \alpha_i = n$, we write $\alpha \vdash n$.

**4.2.** One way to generalize the motivating problem is to fix partitions $\alpha = (\alpha_1, ..., \alpha_p) \vdash n$ and $\beta = (\beta_1, ..., \beta_q) \vdash n$ for some $n \in \mathbb{Z}_{>0}$ and consider two coprime polynomials $P, Q \in \mathbb{C}[X]$ with common degree $\deg(P) = \deg(Q) = n$ such that the multiplicities of the roots of $P$ and $Q$ are determined by the partitions $\alpha$ and $\beta$ respectively. To be more explicit, we consider polynomials $P, Q \in \mathbb{C}[X]$ that can be factorized as

$$P = \prod_{i=1}^{p} (X - a_i)^{\alpha_i}, \quad Q = \prod_{i=1}^{q} (X - b_i)^{\beta_i},$$

and where $a_1, ..., a_p, b_1, ..., b_q$ are the mutually distinct roots of the polynomials $P$ and $Q$. Again, the problem here is to find the minimum possible degree of the difference $R = P - Q$. We will use the setting and notation we have just introduced throughout this whole chapter unless otherwise stated.

## 4.1 Davenport-Zannier polynomials

In this section we will use the Riemann-Hurwitz formula to prove a lower bound for $\deg(R)$ and study conditions to ensure that this bound is actually attained, which will lead to the definition of Davenport-Zannier polynomials.

**Proposition 4.3.** *The degree of the difference $R = P - Q$ satisfies*

$$\deg(R) \geq (n + 1) - (p + q). \tag{11}$$

*Proof.* Let

$$R = P - Q = \prod_{k=1}^{r}(X - c_k)^{\gamma_k}$$

be the factorization of $R$, where $c_1, ..., c_k \in \mathbb{C}$ are its distinct roots with multiplicities $\gamma_1, ..., \gamma_k \ \geq 1$. The degree of $R$ clearly satisfies

$$\deg(R) = \sum_{k=1}^{r} \gamma_k \geq r.$$

The idea of the proof is to use the Riemann-Hurwitz formula for the rational function $f = P/R$ to derive the lower bound (11). By Example 2.22, we know that $f$ defines a holomorphic map

$$f : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C}, \quad x \mapsto \frac{P(x)}{R(x)}$$

from the Riemann sphere to itself, which we also call $f$. We have already seen that the degree of $f$ is $\deg(f) = \max\{\deg(P), \deg(R)\} = n$ and the genus of the Riemann sphere is $g(\mathbb{P}^1\mathbb{C}) = 0$. The only unknown left in the Riemann-Hurwitz formula is the total branching number $b_f$, which can be further expressed in terms of $p, q, r$, and $n$. To do this, we identify possible branch points of $f$.

The preimage of $0 \in \mathbb{P}^1\mathbb{C}$ is $f^{-1}(0) = P^{-1}(0) = \{a_1, ..., a_p\}$. The respective ramification indices are $r_f(a_i) = \alpha_i$ for $i = 1, ..., p$, so 0 is a branch point if and only if there exists an $i = 1, ..., p$ such that $\alpha_i > 1$. To calculate the multiplicities one can use the same approach as we did in Example 3.15. Analogously, for $1 \in \mathbb{P}^1\mathbb{C}$ using the definitions of $f$ and $R$ we get the equivalence

$$f(x) = 1 \iff Q(x) = 0,$$

which implies that the preimage $f^{-1}(1) = \{b_1, ..., b_q\}$ with ramification indices $r_f(b_i) = \beta_i$ for every $i = 1, ..., q$. Hence 1 is a branch point if and only if there exists an $i = 1, ..., q$ such that $\beta_i > 1$.

Next, we look at the preimages of $\infty \in \mathbb{P}^1\mathbb{C}$. Since the leading coefficients of $P$ and $Q$ are 1, it holds $\deg(R) < \deg(P)$ and hence $f(\infty) = \infty$. The only other preimages of $\infty$ are the zeros $c_1, ..., c_r$ of the polynomial $R$. Note that for every $i = 1, ..., r$ we have $P(c_i) \neq 0$, since if it were equal to 0, then $Q(c_i) = (P-R)(c_i) = 0$, which contradicts our assumption that $P$ and $Q$ have no common zeros. The multiplicities are $r_f(c_i) = \gamma_i$ for $i = 1, ..., r$.

Denote the other branch points of $f$ by $y_1, ..., y_m \in \mathbb{P}^1\mathbb{C}$, if there are any. In the case that there are no other branch points, set $m = 0$. For every $i = 1, ..., m$ we write the preimage $f^{-1}(y_i) = \{y'_{i,1}, ..., y'_{i,n_i}\}$, so in particular the point $y_i$ has $n_i$ preimages under the function $f$. Since by assumption $y_i$ is a branch

point, it holds that $n_i < n$ for every $i = 1, ..., m$.
Recall the definition
$$b_f = \sum_{p \in \mathbb{P}^1\mathbb{C}} (r_f(p) - 1)$$
of the total branching number. We can take the sum only over the preimages of the possible branching points we identified above, as all the other points have ramification index 1. Thus

$$\begin{aligned}
b_f = &\sum_{p \in f^{-1}(0)} (r_f(p) - 1) + \sum_{p \in f^{-1}(1)} (r_f(p) - 1) + \sum_{p \in f^{-1}(\infty)} (r_f(p) - 1) + \\
&\sum_{i=1}^{m} \sum_{p \in f^{-1}(y_i)} (r_f(p) - 1) \\
= &\sum_{i=1}^{p} (\alpha_i - 1) + \sum_{i=1}^{q} (\beta_i - 1) + \sum_{i=1}^{r} (\gamma_i - 1) + (r_f(\infty) - 1) + \\
&\sum_{i=1}^{m} \sum_{j=1}^{n_i} (r_f(y'_{i,j}) - 1) \\
= &(n - p) + (n - q) + (n - (r + 1)) + \sum_{i=1}^{m} (n - n_i) \\
= &3n - (p + q + r + 1) + \sum_{i=1}^{m} (n - n_i).
\end{aligned}$$

By substituting the above results in the Riemann-Hurwitz formula and rearranging the terms we get

$$r = n + 1 - (p + q) + \sum_{i=1}^{m} (n - n_i).$$

Since $n_i < n$ for every $i = 1, ..., m$ we conclude that

$$\deg(R) \geq r = (n + 1) - (p + q) + \sum_{i=1}^{m} (n - n_i) \geq (n + 1) - (p + q).$$

□

Our goal is to find pairs of polynomials, such that their difference attains the lower bound in (11), so we introduce the following definition.

**Definition 4.4.** If the difference $R = P - Q$ satisfies

$$\deg(R) = (n + 1) - (p + q),$$

then the pair $(P, Q)$ is called a *Davenport-Zannier-pair* or *DZ-pair*. The pair of partitions $(\alpha, \beta)$ corresponding to the multiplicities of the polynomials $P$ and $Q$ is called the *passport* of the DZ-pair $(P, Q)$.

**4.5.** Looking again at the proof of Proposition 4.3, we can find the following three conditions to ensure that the pair of polynomials $(P, Q)$ is a DZ-pair.
First observe that if $p + q > n + 1$, then the left hand side of (11) is negative, but $\deg(R) \geq 0$. Hence the first condition for the attainability of the lower bound is that $p + q \leq n + 1$.
Secondly, we need to make sure that $\deg(R) = r$. This happens if and only if all the roots of $R$ are simple, or equivalently if and only if the multiplicities satisfy $\gamma_1, ..., \gamma_r = 1$. This can also be expressed in terms of the poles of $f$, which are $\{\infty, c_1, ..., c_r\}$ as seen in the proof. By denoting the ramification index $r_f(\infty) = \gamma_0$, the tuple $\gamma = (\gamma_0, \gamma_1, .., \gamma_r)$ forms a partition of $n$. The second condition for the attainability says that the partition $\gamma \vdash n$ must be of the form $\gamma = (n - r, 1^r)$, where the notation $1^r$ in a tuple means that 1 is repeated $r$ times.
The last condition is that the number $r$ of distinct roots of the polynomial $R$ equals the lower bound $(n+1) - (p+q)$. This happens if and only if the sum $\sum_{i=1}^{m}(n - n_i) = 0$. That is, if there are no other branch points of $f$ except for maybe the points 0, 1, and $\infty$.

## 4.2 Belyĭ functions and dessins d'enfants

The last two conditions we have seen in Paragraph 4.5 above are conditions on the rational function $f = P/R$. In this section we will study functions satisfying these conditions and introduce dessins d'enfants. Further references for the study of dessins d'enfants are the books [JW16, Chapters 1-3], [GG12, Chapter 4]. We use the definitions according to the book [APZ20], which are less general than the ones from [GG12], and [JW16].

**Definition 4.6.** A meromorphic function on the Riemann sphere $\mathbb{P}^1\mathbb{C}$ (or equivalently a rational map between the Riemann sphere and itself) that does not have branch points outside the set $\{0, 1, \infty\}$ is called a *Belyĭ function.*

**Definition 4.7.** Let $f$ be a Belyĭ function. The preimage $D = f^{-1}([0, 1])$ of the unit interval, where preimages of the point 0 are coloured black and preimages of the point 1 are coloured white is called a *dessin* or *dessin d'enfant.*

**4.8.** A dessin is at first a graph drawn on the Riemann sphere. Since the open interval $(0, 1) \subset \mathbb{P}^1\mathbb{C}$ does not contain any branch points of $f$, it gets lifted to $\deg(f)$ homeomorphic copies by $f$. These are the edges of the dessin. In particular, they do not intersect outside of the vertices, which are the preimages of 0 and 1. Hence the dessin corresponds to a plane graph via stereographic projection, and we will usually view it as such. We use the terminology *plane graph* to denote a graph drawn in the plane such that its edges only intersect at the vertices, for an example see Figure 5. Since we will be dealing with graphs, we want to recall some basic definitions. As an example of these definitions on a concrete graph, see again Figure 5.

**Definition 4.9.** A plane graph is *bicoloured* if every vertex has been assigned one out of two colours (we will use white and black) such that every edge connects vertices of different colours.

The dessin of a Belyĭ function $f$ is bicoloured because every edge is a preimage of the unit interval and $f$ is a covering map on the open unit interval. Hence every edge connects a preimage of 0 to a preimage of 1.

**Definition 4.10.** The *degree of a vertex* of a plane graph is the number of edges incident to that vertex.

**Definition 4.11.** The *degree of a face* of a bicoloured plane graph is one half of the number of edges surrounding the face.

Note that the degree of a face is a natural number since in bicoloured plane graphs the number of edges surrounding a face is even. We can also define the degree of a face in analogy to the definition of degree of a vertex as the number of edges incident to the face. For this, we need a suitable definition of what it means for an edge to be incident to a face.

**Definition 4.12.** If we make a circuit around the boundary of a face of a bicoloured plane graph in positive trigonometric (counter clockwise) direction, then an edge is *incident* to that face if we first pass its black vertex and afterwards the white vertex in our circuit. Note that for the outer face the point of view will be "behind" the plane graph.

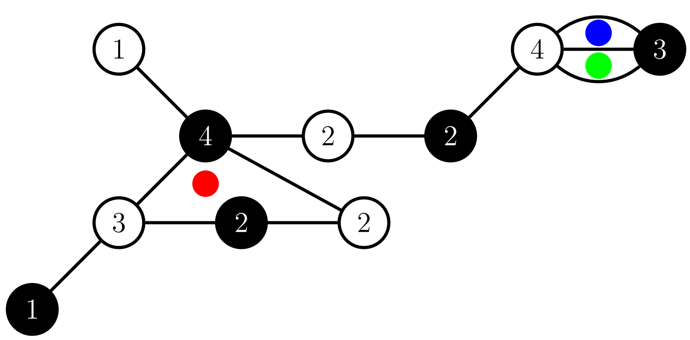



Figure 5: Example of a bicoloured plane graph, the numbers in the vertices are the respective degrees, the faces with the green and blue dot are of degree 1, the face with the red dot is of degree 2, and the outer face is of degree 8

With these definitions, one sees that the number of edges equals the sum of the degrees of the vertices of each colour, and the sum of the face degrees. This holds since each edge is incident to precisely one face, one white, and one black vertex.

**4.13.** A dessin $D$ of a Belyĭ function $f$ has $\deg(f) = n$ edges as $f$ is unramified over the open interval $(0, 1)$.
The degree of a vertex $v$ in $D$ is exactly the ramification index of the corresponding point. This follows from the local representation of $f$ we derived in Proposition 2.24. If $f$ is locally given by the map $z \mapsto z^r$ around $v$, then the open interval $(0, 1)$ lifts to $r$ preimages around $v$. These are exactly the number of edges incident to the vertex $v$. In particular, the degrees of the black vertices are the ramification indices of the roots of $f$, and the degrees of the white vertices are the ramification indices of the preimages of 1 under $f$.
We can also interpret the degrees of the faces in a similar way. Inside each face lies one pole of $f$ and the degree of the face equals the ramification index of that pole. One way to see this is to first consider the preimage of the lines that connect 0 with 1, 1 with $\infty$, and 0 with $\infty$. One can then study, similarly as we did for the preimage of a triangulation, how the preimage of this triangle behaves using the fact that away from the points $0, 1, \infty$ the map $f$ is a covering map of degree $n$. At the points 0, 1, and $\infty$ we understand the map $f$ because of the local representation from Proposition 2.24.

**Definition 4.14.** Let $\alpha = (\alpha_1, ..., \alpha_p), \beta = (\beta_1, ..., \beta_q), \gamma = (\gamma_1, ..., \gamma_r)$ be the respective degrees of the black vertices, white vertices, and faces of a dessin corresponding to a Belyĭ function. The *passport* of the dessin is the triple $(\alpha, \beta, \gamma)$.

**4.15.** For any bicoloured plane graph $G$ we define permutations $x$ and $y$ of the symmetric group $S(E)$ where $E$ is the set of edges in $G$. The permutation $x$ maps an edge $e \in E$ to the edge $xe$ that is the next edge in counterclockwise direction around the black vertex incident to the edge $e$. Analogously, the permutation $y$ sends $e$ to the next edge in counterclockwise direction around the white vertex incident to $e$. For an illustration of the permutations see Figure 6. From the definitions, one sees that the black vertices correspond to the cycles of $x$ and the white vertices to the cycles of $y$. If we first apply the permutation $x$ and then the permutation $y$, we get the permutation $xy$, which can be described as sending an edge $e$ to the next edge incident to the same face as $e$ in clockwise direction. Thus the cycles of $xy$ correspond to the faces of the graph. Note that the cycle length corresponds to the degree of the respective vertex or face.

**Definition 4.16.** The *monodromy group* of the bicoloured plane graph $G$ is the subgroup $\langle x, y \rangle < S(E)$ generated by the permutations $x$ and $y$.

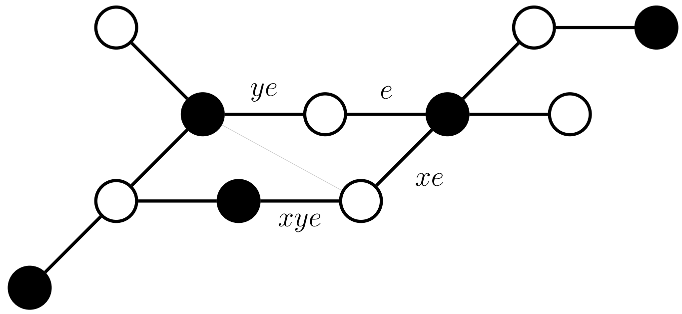



Figure 6: Images of the edge $e$ under the permutations $x$, $y$, and $xy$ as described in 4.15

As the naming might suggest, the monodromy group of a dessin of a Belyĭ function $f$ is equal to the image of the monodromy representation of $f$ according to Definition 2.43. The permutations $x$ and $y$ correspond to the images of the monodromy representation of the positively oriented loops around 0 and 1 respectively. Also note that if the graph $G$ is connected, the monodromy group acts transitively on the set of edges of $G$. We have seen before in 2.42 that the action given by the monodromy representation of $f$ acts transitively as well.
The monodromy group of a connected plane graph $G$ defines $G$ uniquely up to isomorphism. To construct a connected plane graph corresponding to a permutation group $P = \langle x, y \rangle$ acting transitively on a set $E$, one can proceed in the following way: Define an edge for every element in $E$, for each cycle in $x$ and $y$ define a black or white vertex respectively, and for each cycle of $xy$ define a face. Connect an edge to a vertex if the corresponding element in $E$ is in the corresponding cycle of the vertex and analogously an edge is incident to a face if the corresponding element in $E$ is in the cycle corresponding to the face. Since we are only working with plane graphs here, one has to ensure that the permutation data $x, y$ and the set $E$ with $n$ elements correspond to a graph on the plane. The condition

$$\#\{\text{cycles of } x\} + \#\{\text{cycles of } y\} - n + \#\{\text{cycles of } xy\} = 2$$

ensures exactly this. The reason is that the left hand side is the Euler characteristic of the surface in which the graph given by the data can be embedded and so we need it to be equal to $2 = \chi(\mathbb{P}^1\mathbb{C})$ to give a graph in the sphere. For further detail see [GG12, Proposition 4.10]. With this condition, we thus have a correspondence between equivalent plane graphs and the permutation data.

Through this discussion, linking bicoloured plane graphs with Belyĭ functions via the monodromy, one might start to think about a correspondence.

In particular, up to now we have started with a Belyĭ function and arrived at an associated bicoloured plane graph via the dessin of the Belyĭ function. But the following theorem states that is also possible to first consider a bicoloured plane graph, and then associate a Belyĭ function to that graph.

**Theorem 4.17.** *Let $G$ be a bicoloured plane graph, we can view $G$ as a graph drawn on the Riemann sphere. There exists a Belyĭ map $f$ such that its dessin $D = f^{-1}([0,1])$ is isomorphic to $G$ in the following way. There exists an orientation preserving homeomorphism of the Riemann sphere that maps the graph $G$ into $D$ such that the colours of the vertices are preserved. Furthermore, the Belyĭ function $f$ is unique up to a linear fractional transformation.*

A proof of a generalized version of this theorem can be found in [GG12, Chapter 4.2.2], and we will give the general idea of their proof here. One starts by constructing a triangle decomposition associated to the graph on the sphere. This is a decomposition of the sphere into triangles, such that the edges and vertices of the graph belong to intersections of two triangles. As opposed to a triangulation, in a triangle decomposition two triangles are allowed to intersect at more than one edge. The triangles will be constructed such that they belong to one of two types, and at each edge in the graph two triangles of different types meet. Using this triangle decomposition, one can construct a homeomorphism from each triangle to the upper, respective lower halfplane in $\mathbb{C}$ according to the type of the triangle. These homeomorphisms should follow certain restrictions to ensure that they can all be "glued" together to form a continuous function $f : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C}$ such that the restriction

$$f : \mathbb{P}^1\mathbb{C} \setminus f^{-1}\{0,1,\infty\} \to \mathbb{P}^1\mathbb{C} \setminus \{0,1,\infty\}$$

is holomorphic and a topological covering. This can again be extended to a unique holomorphic map $f : \mathbb{P}^1\mathbb{C} \to \mathbb{P}^1\mathbb{C}$ up to linear fraction transformation. The map $f$ only has branch points in $\{0,1,\infty\}$ and thus is a Belyĭ map and its dessin $f^{-1}([0,1]) = G$ equals the graph $G$ we started with. Another approach can be found in [JW16, Chapter 3.3].

If we have a DZ-pair $(P,Q)$, we know from Paragraph 4.5 that the function $f = P/R$ is in fact a Belyĭ function, hence we can consider the dessin of $f$. With the correspondence between Belyĭ functions and bicoloured plane graphs that follows from Theorem 4.17, we also know that given a bicoloured plane graph, there exists a Belyĭ function associated to it. If the Belyĭ function satisfies the conditions we have found in 4.5, we get a DZ-pair. In particular, we can now rewrite the conditions on the Belyĭ function $f$ in 4.5 to conditions on a bicoloured plane graph and get the following proposition.

**Proposition 4.18.** *For $p+q \leq n+1$ the lower bound* $\deg(P-Q) \geq (n+1)-(p+q)$ *derived in Proposition 4.3 is attained if and only if there exists a bicoloured plane graph with the passport $(\alpha, \beta, \gamma)$, for the given partitions $\alpha$ and $\beta$ and where $\gamma$ is the partition $\gamma = (n-r, 1^r)$, here $r = (n+1)-(p+q)$.*

**Remark 4.19.** The partition $\gamma = (n-r, 1^r)$ means geometrically for the plane graph that all faces but the outer one must be of degree 1.

This proposition says that it is enough to show the existence of plane graphs for given partitions to conclude the existence of DZ-pairs. One can show that under a condition on $p$ and $q$, a plane graph as described in Proposition 4.18 exists. If the conditions do not hold, one can show a weaker lower bound. We arrive at the theorem below, which we will not prove here. For a complete proof we refer again to [APZ20]. After stating the general Theorem 4.20, we will prove the solution to our motivating problem, the polynomial Catalan conjecture, in Theorem 4.21.

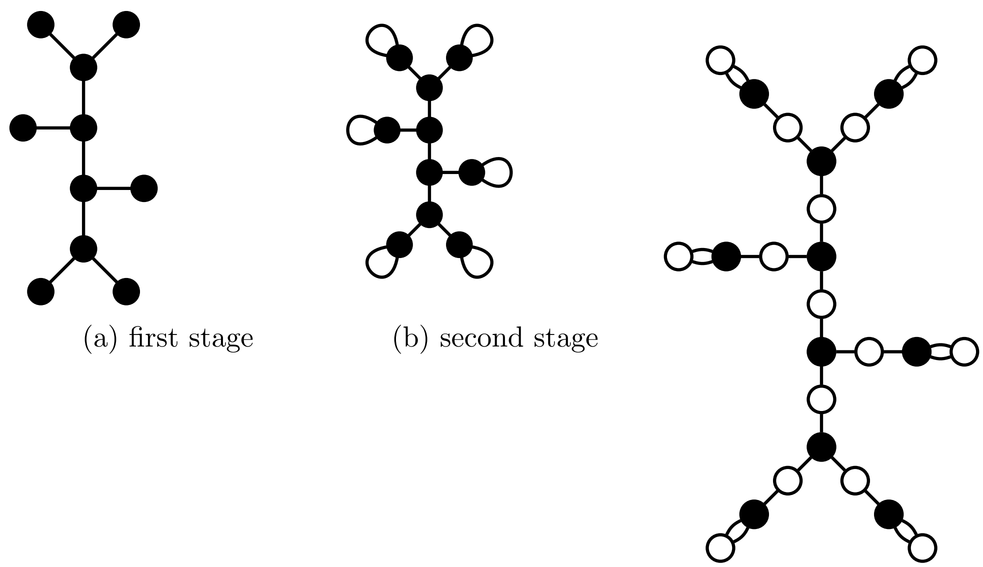
**Theorem 4.20.** *Let $\alpha = (\alpha_1, .., \alpha_p)$ and $\beta = (\beta_1, ..., \beta_q)$ be two partitions of $n$. Let $P, Q$ be two coprime polynomials with complex coefficients and such that the multiplicities of their roots are given by $\alpha$ and $\beta$ respectively, as in 4.2. Denote the difference $P-Q$ by $R$ and the greatest common divisor of $\alpha_1, ..., \alpha_p, \beta_1, ..., \beta_q$ by $d$.*

1. *If $p+q \leq \frac{n}{d}+1$, then* $\deg(R) \geq (n+1)-(p+q)$*. This bound is attained by some polynomials $P$ and $Q$ for any partitions $\alpha, \beta \vdash n$ that satisfy the assumption.*
2. *If $p+q > \frac{n}{d}+1$, then there is a weaker bound* $\deg(R) \geq \frac{(d-1)n}{d}$*. This bound is also attained for some polynomials $P$ and $Q$.*

**Theorem 4.21.** *Let $A, B \in \mathbb{C}[X]$ be coprime polynomials with degrees* $\deg(A) = 2k$ *and* $\deg(B) = 3k$ *for some $k \in \mathbb{Z}_{>0}$, so* $\deg(A^3) = \deg(B^2) = 6k$*. The degree of the difference $R = A^3 - B^2$ satisfies* $\deg(R) \geq k+1$*. Furthermore, this bound is attained for every value of $k$.*

*Proof.* We begin by rewriting the problem such that it fits our situation in 4.2. Consider $P = A^3, Q = B^2$, we have $n = 6k$ and the partitions $\alpha = (3^{2k}), \beta = (2^{3k})$. The number of distinct roots is $p = 2k$ and $q = 3k$ if we assume that the polynomials $A$ and $B$ have simple roots. The lower bound (11) is $(n+1)-(p+q) = 6k+1-(2k+3k) = k+1$. If $A$ or $B$ would have roots of multiplicities larger than 1, then the number $p$ or $q$ would be less than $2k$ or $3k$ respectively. Thus we would not attain the lower bound $k+1$ with this choice of polynomials, therefore we are not concerned with this case.
To show that the lower bound is attained, according to Proposition 4.18, we

need to find a plane graph consisting of $6k$ edges and such that each black vertex has degree 3, each white vertex has degree 2 and each face except the outer one must be of degree 1.
Such plane graphs can be constructed by starting to draw the $2k$ black vertices in such a way that we have $k+1$ outer vertices of degree 1 and all inner vertices are of degree 3. We then add a loop around each outer vertex. Finally, we can add a white vertex into each edge to get the desired plane graph, which proves the attainability of the lower bound $k+1$. For the case $k = 5$, one can see such a plane graph below in Figure 7. $\square$



(a) first stage (b) second stage (c) third stage

Figure 7: Construction of the plane graph that proves the attainability of the lower bound given in Theorem 4.21 for $k = 5$

**Remark 4.22.** We have proven the existence of DZ-pairs for our motivating problem, but to compute concrete examples is another challenge. There is an example from [APZ20, Example 1.2], originally from [E00, Chapter 4], for $k = 5$. The difference of the polynomials

$$\begin{aligned} A^3 &= (X^{10} - 2X^9 + 33X^8 - 12X^7 + 378X^6 + 336X^5 + 2862X^4 + 2652X^3 + \\ &\quad 14397X^2 + 9922X + 18553)^3 \\ B^2 &= (x^{15} - 3x^{14} + 51x^{13} - 67x^{12} + 969x^{11} + 33x^{10} + 10963x^9 + 9729x^8 + \\ &\quad 96507x^7 + 108631x^6 + 580785x^5 + 700503x^4 + 2102099x^3 + \\ &\quad 1877667x^2 + 3904161x + 1164691)^2 \end{aligned}$$

is a polynomial of degree 6, concretely

$$A^3 - B^2 = 2^6 3^{15}(5X^6 - 6X^5 + 111X^4 + 64X^3 + 795X^2 + 1254X + 5477).$$

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